PRACTICE SHEET

1. If, $\vec{a},\vec{b},\vec{c}$ are the position vectors of corners A, B, C of a parallelogram ABCD, then what is the position vector of the corner D?

(a) $\vec{a}+\vec{b}+\vec{c}$ (b) $\vec{a}+\vec{b}-\vec{c}$
(c) $\vec{a}-\vec{b}+\vec{c}$ (d) $-\vec{a}+\vec{b}+\vec{c}$

2. What is the vector in the XY-plane through origin and perpendicular to the vector $\vec{r}=a\hat{i}+b\hat{j}$ and of the same length?

(a) $-a\hat{i}-b\hat{j}$ (b) $a\hat{i}-b\hat{j}$
(c) $-a\hat{i}+b\hat{j}$ (d) $b\hat{i}-a\hat{j}$

3. Given $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$ and $\vec{b}$ is a unit vector co directional with $\vec{a}$. If m is a scalar such that $\vec{b}=m\vec{a}$, then what is the value of m?

(a) 1/5 (b) $1/\sqrt{5}$
(c) 1/29 (d) $1\sqrt{29}$

4. The magnitude of the vectors $\vec{a}$ and $\vec{b}$ are equal and the angle between them is 60° the vectors $\lambda\vec{a}+\vec{b}$ and $\vec{a}-\lambda\vec{b}$ are perpendicular to each other them what is the value of λ?

(a) 1 (b) 2
(c) 3 (d) 4

5. Consider the diagonals of a quadrilateral formed by the vectors $3\hat{i}+6\hat{j}-2\hat{k}$ and $4\hat{i}-\hat{j}+3\hat{k}$. The quadrilateral must be a :

(a) Square (b) Rhombus
(c) Rectangle (d) None of these

6. If $\vec{a}=\hat{i}-2\hat{j}+5\hat{k},\vec{b}=2\hat{i}+\hat{j}-3\hat{k}$ then what is $(\vec{b}.-\vec{a})(3\vec{a}+\vec{b}.)$ equal to?

(a) 106 (b) –106
(c) 53 (d) –53

7. Let $\vec{a},\vec{b}$ and $\vec{c}$ be the position vectors of points A, B and C, respectively. Under which one of the following conditions are the points A, B and C collinear?

(a) $\vec{a}\times\vec{b}=0$
(b) $\vec{b}\times\vec{c}$ is parallel to $\vec{a}\times\vec{b}$
(c) $\vec{a}\times\vec{b}$ is perpendicular to $\vec{b}\times\vec{c}$
(d) $(\vec{a}\times\vec{b})+(\vec{b}\times\vec{c})+(\vec{c}\times\vec{a})=0$

8. If $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-\hat{k}$, then what is $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times(\vec{c}+\vec{a})+\vec{c}\times(\vec{a}+\vec{b})$ is equal to?

(a) $2\hat{i}+3\hat{j}-\hat{k}$ (b) $2\hat{i}-3\hat{j}-\hat{k}$
(c) $3\hat{i}+\hat{j}+\hat{k}$ (d) 0

9. If $(3\vec{a}-\vec{b})\times(\vec{a}+3\vec{b})=k\vec{a}\times\vec{b}$, then what is the value of k?

(a) 10 (b) 5
(c) 8 (d) –8

10. If $\vec{a}$ and $\vec{b}$ are unit vectors, then what is the value of $|\vec{a}\times\vec{b}|^2+(\vec{a}.\vec{b})^2$?

(a) 0 (b) 2
(c) 1 (d) 1/2

11. Let $\vec{a}=(1,-2,3)$ and $\vec{b}=(3,1,2)$ be two vectors and $\vec{c}$ be a vector of length 1 and parallel to $(\vec{a}+\vec{b})$. What is $\vec{c}$ equal to?

(a) $\frac{1}{\sqrt{14}}(-2,-3,1)$ (b) $\frac{1}{\sqrt{2}}(1,0,1)$
(c) $\frac{1}{\sqrt{42}}(-5,-4,-1)$ (d) None of these

12. If $\vec{r}_1=\lambda\hat{i}+2\hat{j}+\hat{k},\vec{r}_2=\hat{i}+(2-\lambda)\hat{j}+2\hat{k}$ are such that $|\vec{r}_1|>|\vec{r}_2|$ then λ satisfies which one of the following?

(a) λ=0 (b) λ=1
(c) λ<1 (d) λ>1

13. If P, Q, R are the mid-points of the sides AB, BC, CA respectively of a ΔABC and $\vec{a},\vec{p},\vec{q}$ are the position vectors of A,P, Q respectively, then what is the position vector of R?

(a) $2\vec{a}-(\vec{p}-\vec{q})$ (b) $(\vec{p}-\vec{q})-2\vec{a}$
(c) $\vec{a}-(\vec{p}-\vec{q})$ (d) $\frac{\vec{a}}{2}-\frac{(\vec{p}-\vec{q})}{2}$

14. If $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors such that $\vec{a}$ is perpendicular to the plane of $\vec{b},\vec{c}$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$.

Then, what is $|\vec{a}+\vec{b}+\vec{c}|$?

(a) 1 (b) 2
(c) 3 (d) 4

15. What is the number of vectors of length 5 units, perpendicular to the vectors $\vec{a}=(1,1,0)$ and $\vec{b}=(0,1,1)$?

(a) 1 (b) 2
(c) 3 (d) 4

16. If $\vec{a}=2\hat{i}-3\hat{j}-\hat{k},\vec{b}=\hat{i}+4\hat{j}-2\hat{k}$, then what is $(\vec{a}+\vec{b})\times(\vec{a}-\vec{b})$ is equal to?

(a) $2(\vec{a}\times\vec{b})$ (b) $-2(\vec{a}\times\vec{b})$
(c) $(\vec{a}\times\vec{b})$ (d) $-(\vec{a}\times\vec{b})$

17. If $\vec{a}$ position vector of a point (1,–3) and A is another point (–1,5) then what are the coordinates of the point B such that $\overline{AB}=\vec{b}$?

(a) (2,0) (b) (0,2)
(c) (–2,0) (d) (0,–2)

18. If $\vec{a}$ is a non-zero vector of modulus a, and λ is a non-zero scalar and $\lambda\vec{a}$ is a unit vector, then:

(a) λ=±1 (b) $\vec{a}=|\lambda|$
(c) $\vec{a}=\frac{1}{|\lambda|}$ (d) $\vec{a}=\frac{1}{\lambda}$

19. Let $\vec{a}$ and $\vec{b}$ be the position vectors of A and B respectively. If C is the point $3\vec{a}-2\vec{b}$, then which one of the following is correct?

(a) C is in between A and B
(b) A is in between C and B
(c) B is in between A and C
(d) A, B and C are not collinear

OFFICIUM SUPREMUM

20. What is the locus of the point (x,y) for which the vectors $\left(\hat{i}-x\hat{j}-2\hat{k}\right)$ and $\left(2\hat{i}+\hat{j}+y\hat{k}\right)$ are orthogonal?

(a) A circle (b) An ellipse

(c) A parabola (d) A straight line

21. If the position vector of a point p with respect to the origin O is $\hat{i}+3\hat{j}-2\hat{k}$ and that of a point Q is $3\hat{i}+\hat{j}-2\hat{k}$, then what is the position vector of the bisector of the $\angle POQ$?

(a) $\hat{i}-\hat{j}-\hat{k}$ (b) $\hat{i}+\hat{j}-\hat{k}$

(c) $\hat{i}+\hat{j}+\hat{k}$ (d) None of these

22. ABCD is a quadrilateral. Force $\overrightarrow{AB}, \overrightarrow{CB}, \overrightarrow{CD}$ and $\overrightarrow{DA}$ act along its sides. What is their resultant?

(a) 2 $\overrightarrow{CD}$ (b) 2 $\overrightarrow{DA}$

(c) 2 $\overrightarrow{BC}$ (d) 2 $\overrightarrow{CB}$

23. PQRS is a parallelogram, where $\overrightarrow{PQ}=3i+2\hat{j}-m\hat{k}$ and $\overrightarrow{PS}=i+3\hat{j}-\hat{k}$ and the area of the parallelogram is $\sqrt{90}$. What is the value of m?

(a) 1 (b) −1

(c) 2 (d) −2

24. What is the value of b such that the scalar product of the vector $\hat{i}+\hat{j}+\hat{k}$ with the unit vector parallel to the sum of the vectors $2\hat{i}+4\hat{j}-5\hat{k}$ and $b\hat{i}+2\hat{j}+3\hat{k}$ is unity?

(a) −2 (b) −1

(c) 0 (d) 1

25. Let a, b and c be the distinct non-negative numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}, c\hat{i}+c\hat{j}+b\hat{k}$ lie on a plane, then which one of the following is correct?

(a) c is the arithmetic mean of a and b

(b) c is the geometric mean of a and b

(c) c is the harmonic mean of a and b

(d) c is equal to zero

26. What is the vector equally inclined to the vectors $\hat{i}+3\hat{j}$ and $3\hat{i}+\hat{j}$?

(a) $\hat{i}+\hat{j}$ (b) $2\hat{i}-\hat{j}$

(c) $2\hat{i}+\hat{j}$ (d) None of these

27. What is the area of a triangle whose vertices are at (3,−1,2), (1, −1, −3) and (4, −3,1) ?

(a) $\frac{\sqrt{165}}{2}$ (b) $\frac{\sqrt{135}}{2}$

(c) 4 (d) 2

28. If $\vec{a}=i-k$, $\vec{b}=xi+j+(1-x)\hat{k}$, $\vec{c}=yi+xj+(1+x-y)\hat{k}$, then $\vec{a}.\left(\vec{b}\times\vec{c}\right)$ depends on

(a) x only (b) y only

(c) Both x and y (d) Neither x nor y

29. Consider the following statements

1. for any three vectors $\vec{a}$ $\vec{b}$ $\vec{c}$

$\vec{a}.\left\{\left(\vec{b}+\vec{c}\right)\times\left(\vec{a}+\vec{b}+\vec{c}\right)\right\}=0$

2. for any three coplanar vectors $\vec{d},\vec{e},\vec{f}$,

$\left(\vec{d}\times\vec{e}\right).\vec{f}=0$

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

30. The vectors $\vec{a}=xi+yj+z\hat{k}$, $\vec{b}=k$, $\vec{c}$ are such that they form a right handed system. What is $\vec{c}$ equal to ?

(a) $\hat{j}$ (b) $y\hat{j}-x\hat{k}$

(c) $y\hat{i}-x\hat{j}$ (d) $x\hat{i}-y\hat{j}$

31. Let $\vec{a}$ and $\vec{b}$ be two unit vectors and α be the angle between them. If ($\vec{a}$ + $\vec{b}$) is also the unit vectors, then what is the value of α ?

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$

(c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{2}$

32. A vector $\vec{b}$ is collinear with the vector $\vec{a}$ = (2,1,−1) and satisfies the condition $\vec{a}.\vec{b}=3$. What is $\vec{b}$ equal to?

(a) (1,1/2,−1/2) (b) (2/3,1/3,−1/3)

(c) (1/2,1/4,−1/4) (d) (1,1,0)

33. What is the value of λ for which the vectors $\hat{i}-\hat{j}+\hat{k}$, $2\hat{i}+\hat{j}-\hat{k}$ and $\lambda\hat{i}-\hat{j}+\lambda\hat{k}$ are coplanar?

(a) 1 (b) 2

(c) 3 (d) 4

34. What is the area of the triangle with vertices (0,2,2), (2,0,−1) and (3,4,0)?

(a) $\frac{15}{2}$ (b) 15

(c) $\frac{7}{2}$ (d) 7

35. If the angle between the vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, then what is the angle between $-5\vec{a}$ and $6\vec{b}$?

(a) $\frac{\pi}{6}$ (b) $\frac{2\pi}{3}$

(c) $\frac{2\pi}{5}$ (d) $\frac{3\pi}{7}$

ANSWER KEYS																			
1.	c	**2.**	d	**3.**	d	**4.**	a	**5.**	b	**6.**	b	**7.**	d	**8.**	d	**9.**	a	**10.**	c
11.	d	**12.**	d	**13.**	c	**14.**	b	**15.**	b	**16.**	b	**17.**	b	**18.**	c	**19.**	b	**20.**	d
21.	a	**22.**	d	**23.**	a	**24.**	d	**25.**	b	**26.**	a	**27.**	a	**28.**	d	**29.**	a	**30.**	c
31.	c	**32.**	a	**33.**	a	**34.**	a	**35.**	b										

Solutions

Sol. 1. (c)
Let O be the origin and ABCD be the parallelogram.
In Δ ODC,
$\overrightarrow{OD} = \overrightarrow{OC} + \overrightarrow{CD}$
$\overrightarrow{CD} = -\overrightarrow{AB}$
and, in ΔAOB,
$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{b} - \vec{a}$
Thus, $\overrightarrow{CD} = -\overrightarrow{AB} = \vec{a} - \vec{b}$
So, $\overrightarrow{OD} = \vec{c} + \vec{a} - \vec{b} \left[\text{sin c}, \overrightarrow{OC} = \vec{C} \text{ and } \overrightarrow{CD} = \vec{a} - \vec{b}\right]$

Sol. 2. (d)
Let $\vec{r_1} = b\hat{i} - a\hat{j}$ be the required vector
Given $\vec{r} = a\hat{i} - b\hat{j}$
Now, $\vec{r_1}.\vec{r} = (b\hat{i} - a\hat{j}).(a\hat{i} + b\hat{j})$
$= ab - ab = 0$
Hence, option (d) is the correct answer

Sol. 3. (d)
Given $\hat{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$
Also, $\hat{b} = m\vec{a} = m(2\hat{i} - 3\hat{j} + 4\hat{k})$
As $\hat{b}$ is a unit vector therefore $|\hat{b}| = 1$
Now, $|2\hat{i} - 3\hat{j} + 4\hat{k}| = \sqrt{4+9+16} = \sqrt{29}$
Therefore, m should be $\frac{1}{\sqrt{29}}$

Sol. 4. (a)
Since, vectors $\lambda\vec{a} + \vec{b}$ and $a - \lambda\vec{b}$ are perpendicular to each other therefore
$(\lambda\vec{a} + \vec{b}).(\vec{a} - \lambda\vec{b}) = 0$
$\Rightarrow \lambda\vec{a}.\vec{a} - \lambda^2\vec{a}\vec{b} + \vec{b}.\vec{a} - \lambda\vec{b}.\vec{b} = 0$
$\Rightarrow \lambda|\vec{a}|^2 + (1-\lambda^2)\vec{a}.\vec{b} - \lambda|\vec{b}|^2 = 0$
$\Rightarrow \lambda|\vec{a}|^2 + (1-\lambda^2)\vec{a}.\vec{b} - \lambda|\vec{a}|^2 = 0 \left(\because |\vec{a}| = |\vec{b}|\right)$
$\Rightarrow (1-\lambda^2)\,\vec{a}.\vec{b}$
Since, $\cos 60° = \frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{a}|} = \frac{\vec{a}.\vec{b}}{|\vec{a}|^2} = \vec{a}.\vec{b}$
$\therefore (1-\lambda^2)\,\vec{a}.\vec{b}\ (1-\lambda^2)\cos 60°$
$\therefore (1-\lambda^2)1/2 = 0 \Rightarrow \lambda = \pm 1$

Sol. 5. (b)
Let $\vec{d_1}$ and $\vec{d_2}$ be the two diagonals of a quadrilateral such that
$\vec{d_1} = 3\hat{i} + 6\hat{j} - 2\hat{k}$
and $\vec{d_2} = 4\hat{i} - \hat{j} + 3\hat{k}$
now, Dot product of $\vec{d_1}$ and $\vec{d_2}$ is
$\vec{d_1}.\vec{d_2} = 3(4) + 6(-1) - 2(3) = 0$
Now, $|\vec{d_1}| = \sqrt{3^2 + 6^2 + 2^2} = 7$
$|\vec{d_2}| = \sqrt{4^2 + 1^2 + 3^2} = \sqrt{26}$
Since, $|\vec{d_1}| \neq |\vec{d_2}|$
Hence given quadrilateral is a rhombus,

Sol. 6. (b)
Given, $\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$
and $\vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$
$\therefore \vec{b} - \vec{a} = 2\hat{i} + \hat{j} - 3\hat{k} - \hat{i} + 2\hat{j} - 5\hat{k} = \hat{i} + 3\hat{j} - 8\hat{k}$
and $(3\vec{a} + \vec{b}) = (3\hat{i} - 6\hat{j} + 15\hat{k}) + 2\hat{i} + \hat{j} - 3\hat{k}$
$= 5\hat{i} - 5\hat{j} + 12\hat{k}$
Hence, $(\vec{b} - \vec{a})(3\vec{a} + \vec{b}) = (\hat{i} + 3\hat{j} - 8\hat{k})(5\hat{i} - 5\hat{j} + 1)$
$= 5 - 15 - 106$
$= -106$

Sol. 7. (d)
Points A, B and C are collinear, if
$(\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) = 0$

Sol. 8. (d)
Since, $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - \hat{k}$
$\Rightarrow \vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b})$
$= (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c}) + (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{a}) + (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{b})$
$= (\vec{a} \times \vec{b}) - (\vec{c} \times \vec{a}) + (\vec{b} \times \vec{c}) - (\vec{a} \times \vec{b}) + (\vec{c} \times \vec{a}) - (\vec{b} \times \vec{c}) = 0$

Sol. 9. (a)
$(3\vec{a} - \vec{b}) \times (\vec{a} + 3\vec{b})$
$= (3\vec{a} - \vec{b}) \times \vec{a} + (3\vec{a} - \vec{b}) \times 3\vec{b}$
$= 3\vec{a} \times \vec{b} - \vec{b} \times \vec{a} + 3\vec{a} \times 3\vec{b} - \vec{b} \times 3\vec{b}$
$= 0 - (-\vec{a} \times \vec{b}) + 9\vec{a} \times \vec{b} - 0$
$= 10\,\vec{a} \times \vec{b}$
$\therefore k = 10$

Sol. 10. (c)
$|\vec{a} \times \vec{b}|^2 + (\vec{a}.\vec{b})^2$
$= (|\vec{a}|.|\vec{b}|\sin\theta)^2 (|\vec{a}|.|\vec{b}|\cos\theta)^2$
$= (1.1.\sin\theta)^2 + (1.1.\cos\theta)^2$
$= \sin^2\theta + \cos^2\theta = 1$

Sol. 11. (d)
Given $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$
and $\vec{b} = 3\hat{i} + \hat{j} + 2\hat{k}$
$\therefore \vec{a} + \vec{b} = 4\hat{i} - \hat{j} + 5\hat{k}$
Then, $\vec{c} = \lambda(\vec{a} + \vec{b})$
$= \lambda\,(4\hat{i} - \hat{j} + 5\hat{k})$
$\Rightarrow l = \sqrt{16\lambda^2 + \lambda^2 + 25\lambda^2}$
$\Rightarrow l = \sqrt{42}\lambda$
$\lambda \square \frac{l}{\sqrt{42}}$
$\therefore \vec{c} = \frac{l}{\sqrt{42}}(4\hat{i} + \hat{j} + 5\hat{k}) = \frac{l}{\sqrt{42}}(4,-1,5)$

Sol. 12. (d)
Given $\vec{r_1} = \lambda\hat{i} - 2\hat{j} + 3\hat{k}$
and $\vec{r_1} = \hat{i} + (2-\lambda)\hat{j} + 2\hat{k}$
$\therefore |\vec{r_1}| > |\vec{r_2}|$
$\Rightarrow \sqrt{\lambda^2 + (2)^2 + (1)^2} > \sqrt{(1)^2 + (2-\lambda)^2 + (2)^2}$
$\Rightarrow \lambda^2 + 4 + 1 > 1 + 4 + \lambda^2 - 4\lambda + 4$
$\Rightarrow 5 > 9 - 4\lambda$
$\Rightarrow 4\lambda > 4$
$\Rightarrow \lambda > 1$

Sol. 13. (c)
Let the position vectors of B,C and R are $\vec{b}, \vec{c}$ and $\vec{r}$ respectively.
$\therefore \vec{p} = \frac{\vec{a} + \vec{b}}{2} \Rightarrow \vec{b} = 2\vec{p} - \vec{a}$
$\vec{q} = \frac{\vec{b} + \vec{c}}{2}$
$\Rightarrow \vec{c} = 2\vec{q} - \vec{b}$
$\Rightarrow \vec{c} = 2\vec{q} - (2\vec{p} - \vec{a}) = 2\vec{q} - 2\vec{p} + \vec{a}$
and $\vec{r} = \frac{\vec{a} + \vec{c}}{2}$
$= \frac{2\vec{q} - 2\vec{p} + \vec{a} + \vec{a}}{2} = \vec{q} - \vec{p} + \vec{a} = \vec{a} - (\vec{p} - \vec{q})$

Sol. 14. (b)
As given: $\vec{a}$ is perpendicular to $\vec{b}$ and $\vec{c}$
$\Rightarrow \vec{a}.\vec{b} = 0\ \&\ \vec{a}.\vec{c} = 0$ and angle between
$\vec{b}$ and $\vec{c} = \frac{}{3}$
$\therefore \vec{b}.\vec{c} = |\vec{c}||\vec{c}|\cos\frac{\pi}{3} = 1.1\frac{1}{2}$
$= \frac{1}{2}\left(\because \vec{b} \text{ and } \vec{c} \text{ are unit vectors}\right)$
Now,
$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a})$
$= 1+1+1+2.\left(0 + \frac{1}{2} + 0\right)$
$= 1+1+1+1 = 4$
$\Rightarrow |\vec{a} + \vec{b} + \vec{c}| = 2$

Sol. 15. (b)
As given, vectors are: $a = \hat{i} + \hat{j}$ and $b = \hat{j} + \hat{k}$
So, $\vec{a} \times \vec{b}$ is perpendicular to $\vec{a}$ and $\vec{b}$
$$a \times b = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$
$= = \hat{i} - \hat{j} - \hat{k}$
and $|\vec{a} \times \vec{b}| = \sqrt{1+1+1} = \sqrt{3}$
Unit vector perpendicular to $\vec{a}$ and $\vec{b}$

$= \pm \frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|} = \frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$

Thus, the number of vectors perpendicular to the vectors $\vec{a}$ and $\vec{b}$ is 2. This is true for vectors of any length. So, it is true for vector of length 5 unit.

Sol. 16. (b)

Given vectors are

$\vec{a} = 2\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 4\hat{j} - 2\hat{k}$

$\Rightarrow \vec{a}+\vec{b} = (2\hat{i}-3\hat{j}-\hat{k}) + (\hat{i}+4\hat{j}-2\hat{k})$

$= 3\hat{i}+\hat{j}-3\hat{k}$

and $\vec{a}-\vec{b} = (2\hat{i}-3\hat{j}-\hat{k}) + (\hat{i}+4\hat{j}-2\hat{k})$

$= \hat{i}-7\hat{j}+\hat{k}$

$\therefore (\vec{a}+\vec{b})\times(\vec{a}-\vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -3 \\ 1 & -7 & 1 \end{vmatrix}$

$= \hat{i}\begin{vmatrix} 1 & -3 \\ -7 & 1 \end{vmatrix} \hat{j}\begin{vmatrix} 3 & -3 \\ 1 & 1 \end{vmatrix} \hat{k}\begin{vmatrix} 3 & 1 \\ 1 & -7 \end{vmatrix}$

$= \hat{i}(1-21) - \hat{j}(3+3) + \hat{k}(-21-1)$

$= -20\hat{i} - 6\hat{j} - 22\hat{k}$

$= -2(10\hat{i}+3\hat{j}+11\hat{k})$

Now, $\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & -1 \\ 1 & 4 & -2 \end{vmatrix}$

$\hat{i}\begin{vmatrix} -3 & -1 \\ 4 & -2 \end{vmatrix} - \hat{j}\begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} + \hat{k}\begin{vmatrix} 2 & -3 \\ 1 & 4 \end{vmatrix}$

$= \hat{i}(6+4) - \hat{j}(-4+1) + \hat{k}(8+3)$

$= 10\hat{i} - 3\hat{j} - 11\hat{k}$

Hence, $(\vec{a}+\vec{b})\times(\vec{a}-\vec{b}) = -2(\vec{a}\times\vec{b})$

Sol. 17. (b)

Let the coordinates of B be (x, y).

$\vec{a} = \hat{i} - 3\hat{j}$

P.V. of A is (–1, 5) so,

$\overrightarrow{OA} = \hat{i}+5\hat{j}, \overrightarrow{OB} = x\hat{i}+y\hat{i}$

$\therefore \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{a}$

$\Rightarrow (x+1)\hat{i} + (y-5)\hat{j} = \hat{i} - 3\hat{j}$

$\Rightarrow x + 1 = 1$ and $y - 5 = -3$

$\Rightarrow x = 0$ and $y = 2$

$\therefore$ Coordinates of B are (0,2)

Sol. 18. (c)

As given, $\lambda\vec{a}$ is a unit vector.

$\Rightarrow |\lambda\vec{a}| = 1$

$\Rightarrow |\lambda||\vec{a}| = 1$

$\Rightarrow a = \frac{1}{|\lambda|} \quad [\because |\vec{a}| = a]$

Sol. 19. (b)

As given : $\bar{a}$ and $\bar{b}$ are position vectors of A and B respectively and position vector of C is $3\vec{a} - 2\vec{b}$

$\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$, where O is the origin and

$\overrightarrow{OC} = 3\vec{a} - 2\vec{b}, \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{b} - \vec{a}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = 3\vec{a} - 2\vec{b} - \vec{a} = 2\vec{a} - 2\vec{b}$

$\Rightarrow \overrightarrow{AC} = 2(\vec{a}-\vec{b}) = -2(\vec{b}-\vec{a}) = -1 = -2\overrightarrow{AB}$

So, $\overrightarrow{AC}$ is opposite to $\overrightarrow{AB}$ so

A is between C and B and position vector of C shows an external division by C.

Sol. 20. (d)

As given: $(\hat{i} - x\hat{j} - 2\hat{k})$ and $(2\hat{i} + \hat{j} - y\hat{k})$ are orthogonal

So, there dot product = 0

$\Rightarrow (\hat{i} - x\hat{j} - 2\hat{k}).(2\hat{i} + \hat{j} + y\hat{k}) = 0$

$\Rightarrow 2 - x - 2y = 0$

$\Rightarrow x + 2y = 2$

Which is an equation of straight line.

Thus, the locus of the point (x, y) is a straight line.

Sol. 21. (a)

Let $\overrightarrow{OP} = i + 3\hat{j} - 2\hat{k}$ and $\overrightarrow{OQ} = 3i + \hat{j} - 2\hat{k}$

Let $\hat{i} + \hat{j} - \hat{k}$ be required position vector of the bisector of the angle POQ since, it is the bisector of $\angle POQ$ therefore. It will make equal angle s with $\overrightarrow{OP}$ and $\overrightarrow{OQ}$.

Let Angle between $\hat{i} + 3\hat{j} - 2\hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$ is

$\theta = \cos^{-1}\left(\frac{1+3+2}{\sqrt{1+9+4}\sqrt{1+1+1}}\right)$

$= \cos^{-1}\left(\frac{6}{\sqrt{14}\sqrt{3}}\right)$

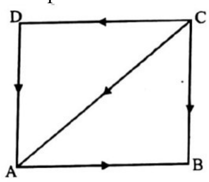
and angle between $3\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$, is

$\phi = \cos^{-1}\left(\frac{1+2+2}{\sqrt{9+1+4}\sqrt{1+1+1}}\right) = \cos^{-1}\left(\frac{6}{\sqrt{4}\sqrt{3}}\right)$

Hence, $\theta = \phi$

Sol. 22. (d)

Let ABCD be a quadrilateral



$\overrightarrow{AB} + \overrightarrow{CB} + \overrightarrow{CD} + \overrightarrow{DA}$

$= \overrightarrow{AB} + \overrightarrow{CB} + \overrightarrow{CA} (\because \overrightarrow{} \quad \overrightarrow{A} = \overrightarrow{CA})$

$= \overrightarrow{AB} + \overrightarrow{CA} + \overrightarrow{CB}$

$= 2\overrightarrow{CB}$

Sol. 23. (a)

Let $\overrightarrow{PQ} = 3i + 2\hat{j} - m\hat{k}$ and $\overrightarrow{PS} = i + 3\hat{j} - \hat{k}$

where PQRS is a parallelogram

$\therefore$ Area of parallelogram $= \left\|\begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & -m \\ 1 & 3 & 1 \end{matrix}\right\|$

$= \left|\hat{i}(2+3m) - \hat{j}(3+m) + \hat{k}(9-2)\right|$

$= \sqrt{(2+3m)^2 + (3+3m)^2 + 7^2}$

$\Rightarrow 90 = 4 + 9m^2 + 12m + 9 + m^2 + 6m + 49$

$\Rightarrow 10m^2 + 18m - 28 = 0$

$\Rightarrow 5m^2 + 9m - 14 = 0$

$\Rightarrow 5m^2 + 14m - 5m - 14 = 0$

$\Rightarrow (5m+14)(m-1) = 0$

$\Rightarrow m = 1 \, or \, \frac{-14}{5}$

Sol. 24. (d)

Let $\vec{A} = i + j + k, \vec{B} = 2i + 4\hat{j} - 5\hat{k}$ and $\vec{C} = bi + 2\hat{j} + 3\hat{k}$

Now, $\vec{B} + \vec{C} = 2i + 4\hat{j} - 5\hat{k} + b\hat{i} + 2\hat{j} + 3\hat{k}$

$= (2+b)\hat{i} + 6\hat{j} - 2\hat{k}$

The unit vector parallel to

$\vec{B} + \vec{C} = 2i + 4\hat{i} - 5\hat{k} + b\hat{i} + 2\hat{j} + 3\hat{k}$

$= (2+b)\hat{i} + 6\hat{j} - 2\hat{k}$

The unit vector parallel to $\vec{B} + \vec{C}$ is

$\hat{n} = \frac{(2+b)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{(2+b)^2 + 6^2 + (-2)^2}} =$

$\hat{n} = \frac{(2+b)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{b^2 + 4b + 44}}$

Now, $(\hat{i} + \hat{j} + \hat{k}).\hat{n} = 1 \Rightarrow \frac{2+b+6-2}{\sqrt{b^2+4b+44}} = 1$

$2 + b + 6 - 2 = \sqrt{b^2 + 4b + 44}$

$\Rightarrow 8b = 8$

$\Rightarrow b = 1$

Sol. 25. (b)

Given $a\hat{i} + a\hat{j} + c\hat{k}, \hat{i} + \hat{k}.$ and $c\hat{i} + c\hat{j} + b\hat{k}$ lie on a plane.

$\Rightarrow$ vector $a\hat{i} + a\hat{j} + c\hat{k}, \hat{i} + \hat{k}.$ and $c\hat{i} + c\hat{j} + b\hat{k}$ are coplanar

$\therefore \begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$

$\Rightarrow a(-c) - a(b-c) + c(c) = 0$

$\Rightarrow -ac - ab + ac + c^2 = 0$

$\Rightarrow c^2 = ab$

$\Rightarrow c$ is the geometric mean of a and b

Sol. 26. (a)

Let the required vector be $\hat{i} + \hat{j}$

Since the vector $\hat{i} + \hat{j}$ is equally inclined to the vectors $\hat{i} + 3\hat{j}$ and $3\hat{i} + \hat{j}$ therefore angle b/w $\hat{i} + \hat{j}$ and $3\hat{i} + \hat{j} = \theta_2$

$\therefore$ Angle between $\hat{i} + \hat{j}$ and $\hat{i} + 3\hat{j}$

$= \cos^{-1}\left[\frac{(1)(1)+(1)(3)}{\sqrt{(1)^2+(1)^2}\sqrt{(1)^2+(3)^2}}\right]$

$= \cos^{-1}\left[\frac{1+3}{\sqrt{2}\sqrt{10}}\right] = \cos^{-1}\left[\frac{4}{\sqrt{2}\sqrt{10}}\right]$

$= \cos^{-1}\left[\frac{2}{\sqrt{5}}\right]$ and

Angle between $\hat{i} + \hat{j}$ and $(3\hat{i} + \hat{j})$

$= \cos^{-1}\left[\frac{1+3}{\sqrt{10}\sqrt{2}}\right]$

$= \cos^{-1}\left(\frac{1+3}{\sqrt{2}\sqrt{10}}\right) = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Sol. 27. (a)

Let the vertices of the ΔABC are A $(3,-1,2)$, $B(1,-1,-3)$ $C(4,-3,1)$

Let $\overrightarrow{OA} = 3i - \hat{j} + 2\hat{k}$

$\overrightarrow{OB} = i - \hat{j} + 3\hat{k}$ and

$\overrightarrow{OC} = 4i - 3\hat{j} + k$

Area of $\Delta ABC = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right|$

Now, $\overrightarrow{AB} = \overrightarrow{OA} - \overrightarrow{OB} = 2i + 5\hat{k}$

$\overrightarrow{AC} = \overrightarrow{OA} - \overrightarrow{OC} = i + 2\hat{j} + \hat{k}$

$\therefore$ Required Area $= \frac{1}{2}\left\|\begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 5 \\ -1 & 2 & 1 \end{matrix}\right\|$

$= \frac{1}{2}\left|\hat{i}(-10) - \hat{j}(2+5) + \hat{k}(4)\right|$

$= \frac{1}{2}\left|-10\hat{i} - 7\hat{j} + 4\hat{k}\right|$

$= \frac{1}{2}\sqrt{100+49+16} = \frac{1}{2}\sqrt{165} sq.unit$

Sol. 28. (d)

Let $\vec{a} = i - \vec{k}$, $\vec{b} = xi + j + (1-x)\hat{k}$

and $\vec{c} = yi + xj + (1+x-y)\hat{k}$

Now, $(\vec{b} \times \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & 1 & (1-x) \\ y & x & (1+x-y) \end{vmatrix}$

$= \hat{i}(1+x-y-x+x^2) -$

$\hat{j}(x+x^2-xy-y) + \hat{k}(x^2-y)$

$\hat{i}(1-y+x^2) - \hat{j}(x-x^2-xy-y) +$

$\hat{k}(x^2-y)$

Now, $\vec{a}.(\vec{b} \times \vec{c}) = 1(1-y+x^2) + 0$

$(x+x^2-xy-y) - 1(x^2-y)$

$= 1 - y + x^2 - x^2 + y$

=1 which that $\vec{a}.(\vec{b} \times \vec{c})$ does not depend on x and y.

Sol. 29. (a)

Consider statement 1

$\vec{a}.\{(\vec{b}+\vec{c}) \times (\vec{a}+\vec{b}+\vec{c})\} = 0$

$= \vec{a}.\{\vec{b}\times\vec{a} + \vec{b}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a} + \vec{c}\times\vec{b} + \vec{c}\times\vec{c}\}$

$= 0 + 0 + \vec{a}.(b \times \vec{c}) + 0 + \vec{a}.(c \times \vec{b}) + 0$

$(\because \quad))$

and for any three coplanar vector $\vec{d}, \vec{e}, \vec{f}$,

$(\vec{d} \times \vec{e}).\vec{f} = 0$

Hence, statement (1) is correct and statement-2 is incorrect.

Sol. 30. (c)

We know, scalar triple product $(\vec{a} \times \vec{b}).\vec{c}$ is positive or negative according as $\vec{a}$, $\vec{b}$, $\vec{c}$ form a right handed or consider option (a)

Let $\vec{c} = \vec{j}$

$\therefore \left[\vec{a}\ \vec{b}\ \vec{c}\right] = \begin{vmatrix} x & y & z \\ 0 & 0 & 1 \\ 0 & y-x \end{vmatrix} = x(-y) - y(0) + z(0) = -xy$

Option (b)

Let $\vec{c} = y\ j - x\ \hat{k}$

$\therefore \left[\vec{a}\vec{b}\vec{c}\right] = \begin{vmatrix} x & y & z \\ 0 & 0 & 1 \\ 0 & 0 & -x \end{vmatrix} = x(-y) - y(0) + z(0)$

$= -xy$

Option (C)

Let $\vec{c} = yi - x\ \hat{j}$

$\therefore \left[\vec{a}\vec{b}\vec{c}\right] = \begin{vmatrix} x & y & z \\ 0 & 0 & 1 \\ y & -x & 0 \end{vmatrix} = x(x) - y(-y) + z(0)$

$= x^2 + y^2$

Since, scalar triple product is positive when $\vec{c} = y\ i - x\ \hat{j}$

$\therefore$option (c) is correct

Sol. 31. (c)

Let $\vec{a}$ and $\vec{b}$ be two unit vectors

$\therefore \ |a| = 1$ and $|b| = 1$

Since,α is the angle between $\vec{\alpha}$ and $\vec{b}$

$\therefore \cos\alpha = \frac{\vec{a}.\vec{b}}{|a||b|}$

$\cos\alpha = \frac{\vec{a}.\vec{b}}{1}$

$\cos\alpha = \vec{a}.\vec{b}$

Now, $\left|\vec{a} + \vec{b}\right| = 1(\because \quad$ unit vector$)$

Squaring both sides

$|\vec{a}| + |\vec{b}| + 2\vec{a}.\vec{b} = 1$

$\Rightarrow 1 + 1 + 2\cos\ \alpha = 1$

$\Rightarrow 2\cos\ \alpha = -1$

$\Rightarrow \cos\ \alpha = -\frac{1}{2} = \cos\frac{2\pi}{3}$

$\Rightarrow \alpha = \frac{2\pi}{3}$

Sol. 32. (a)

Let $\vec{b} = x\vec{i} + y\ \vec{j} + z\ \vec{k}$

Since, $\vec{b}$ is collinear with vector $\vec{a}$ therefore $\vec{a} = k\ \vec{b}$ where k is a scalar.

Given $\vec{a}$ =2,1,-1

$\therefore (2,1,-1) = k(x,y,z)$

$\Rightarrow x = \frac{2}{k},\ y\frac{1}{k},\ z = \frac{-1}{k}$

Also, $\vec{a}.\vec{b} = 3$

$\Rightarrow 2x + y - z = 3$

$\Rightarrow 2\left(\frac{2}{k}\right) + \frac{1}{k} + \frac{1}{k} = 3$

$\Rightarrow \ \frac{4}{k} + \frac{1}{k} + \frac{1}{k} = 3$

$\Rightarrow \ \frac{6}{k} = 3 \Rightarrow k = 2$

$\therefore x = 1,\ y = \frac{1}{2}$ and z=$\frac{-1}{2}$

Hence $\vec{b} = \left(1, \frac{1}{2}, \frac{-1}{2}\right)$

Sol. 33. (a)

Given vectors are

$\hat{i} - \hat{j} + \hat{k},\ 2\hat{i} + \hat{j} - \hat{k}$ and $\lambda\hat{i} - \hat{j} + \lambda\hat{k}$

We know given vectors are coplanar, if

$\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & -1 \\ \lambda & -1 & \lambda \end{vmatrix} = 0$

$\Rightarrow 1(\lambda - 1) + 1(2\lambda + \lambda) + 1(-2 - \lambda) = 0$

$\Rightarrow \lambda - 1 + 3\lambda - 2 - \lambda = 0$

$\Rightarrow 3\lambda = 3 \Rightarrow \lambda = 1$

Sol. 34. (a)

Let $A = (0,2,2), B = (2,0-1)$ and $C = (3,4,0)$

$\overrightarrow{AB} = (2-0, 0-2-1-2)$ and

$\overrightarrow{AC} = (3-0, 4-2, 0-2)$

$\Rightarrow \overrightarrow{AB} = (2,-2,-3)$ and $\overrightarrow{AC} = \overrightarrow{AC} = (3,2,-2)$

$\therefore$ Area of triangle

$= \frac{1}{2} \times$ magnitude of $\overrightarrow{AB} \times \overrightarrow{AC}$

$= \frac{1}{2}\left\|\overrightarrow{AB} \times \overrightarrow{AC}\right\|$

$= \frac{1}{2}\left\|\begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & -3 \\ 3 & 2 & -2 \end{matrix}\right\|$

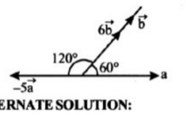
$= \frac{1}{2}\left[\left[i(4+6) - \hat{j}(-4+9) + \hat{k}(4+6)\right]\right]$

$= \frac{1}{2}\left|10\hat{i} - 5\hat{j} + 10\hat{k}\right|$

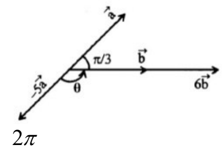
$= \frac{1}{2}\sqrt{(10)^2 + (5)^2 + (10)^2} = \frac{1}{2}\sqrt{225} = \frac{15}{2}$

Sol. 35. (b)

From the figure it is clear that the angle between $6\vec{b}$ and $-5\vec{a}$ is 120^0 or $\frac{2\pi}{3}$



ALTERNATE SOLUTION:



$\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

NDA PYQ

1. If $x\hat{i}+y\hat{j}+z\hat{k}$ is a unit vector and x : y : z = $\sqrt{3}:2:3$ then what is the value of z?
(a) 3/16 (b) 3
(c) 3/4 (d) 2
[NDA (I) - 2011]

2. What is the projection of the vector $\hat{i}-2\hat{j}+\hat{k}$ on the vector $4\hat{i}-4\hat{j}+7\hat{k}$?
(a) $\frac{\sqrt{5}}{2}$ (b) $\frac{19}{9}$
(c) $\frac{\sqrt{5}}{4}$ (d) $\frac{11}{3}$
[NDA (I) - 2011]

3. If the vector $\vec{a}$ lies in the plane of vectors $\vec{b}$ and $\vec{c}$, then which one of the following is correct?
(a) $\vec{a}.(\vec{b}\times\vec{c})=0$ (b) $\vec{a}.(\vec{b}\times\vec{c})=1$
(c) $\vec{a}.(\vec{b}\times\vec{c})=-1$ (d) $\vec{a}.(\vec{b}\times\vec{c})=3$
[NDA (I) - 2011]

4. If p,q,r and s be respectively the magnitude of the vectors $3\hat{i}-2\hat{j}$, $2\hat{i}+2\hat{j}+\hat{k}$, $4\hat{i}-\hat{j}+\hat{k}$, $2\hat{i}-2\hat{j}+3\hat{k}$ Then, which one of the following is correct?
(a) r > s > q > p (b) s > r > p > q
(c) r > s > p > q (d) s > r > q > p
[NDA (I) - 2011]

5. Which one of the following is the unit vector perpendicular to the vectors $4\hat{i}+2\hat{j}$ and $-3\hat{i}+2\hat{j}$?
(a) $\frac{\hat{i}+\hat{j}}{2}$ (b) $\frac{\hat{i}-\hat{j}}{\sqrt{2}}$
(c) $\hat{k}$ (d) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{2}}$
[NDA (I) - 2011]

6. Consider the following statements in respect of the vectors $\overrightarrow{u_1}=(1,2,3)$, $\overrightarrow{u_2}=(2,3,1)$, $\overrightarrow{u_3}=(1,3,2)$ and $\overrightarrow{u_4}=(4,6,2)$
I. $\overrightarrow{u_1}$ is parallel to $\overrightarrow{u_4}$
II. $\overrightarrow{u_2}$ parallel to $\overrightarrow{u_4}$
III $\overrightarrow{u_3}$ is parallel to $\overrightarrow{u_4}$
Which of the statements given above are correct?
(a) only I (b) only II
(c) only III (d) Both I and III
[NDA (I) - 2011]

7. The point with position vectors $10\hat{i}+3\hat{j}, 12\hat{i}-5\hat{j}, a\hat{i}+11\hat{j}$ are collinear, if the value of a is:
(a) –8 (b) 4
(c) 8 (d) 12
[NDA (I) - 2011]

8. What is the sine of angle between the vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+2\hat{j}+3\hat{k}$?
(a) $\sqrt{\frac{13}{7}}$ (b) $\frac{\sqrt{13}}{7}$
(c) $\frac{13}{\sqrt{7}}$ (d) none of the above
[NDA (I) - 2011]

9. What is the area of the triangle with vertices (1,2,3) , (2,5,−1) , and (−1,1,2) ?
(a) $\frac{\sqrt{155}}{2}$ (b) $\frac{\sqrt{175}}{2}$
(c) $\frac{\sqrt{155}}{4}$ (d) $\frac{\sqrt{175}}{4}$
NDA (II) - 2011]

10. If $\vec{c}$ is the unit vector perpendicular to both the vector $\vec{a}$ and $\vec{b}$, then what is another unit vector perpendicular to both the vectors $\vec{a}$ and $\vec{b}$?
(a) $\vec{c}\times\vec{a}$ (b) $\vec{c}\times\vec{b}$
(c) $-\frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$ (d) $\frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$
[NDA (II) - 2011]

11. If the vectors $-\hat{i}-2x\hat{j}-3y\hat{k}$ and $\hat{i}-3x\hat{j}-2y\hat{k}$ are orthogonal to each other, then what is the locus of the point (x,y) ?
(a) a straight line (b) an ellipse
(c) a parabola (d) circle
[NDA (II) - 2011]

12. For what value of m is the point with position vectors $10\hat{i}+3\hat{j}, 12\hat{i}-5\hat{j}$ and $m\hat{i}+11\hat{j}$ collinear?
(a) –8 (b) 4
(c) 8 (d) 12
[NDA (II) - 2011]

13. For what value of m are the vector $2\hat{i}-3\hat{j}+4\hat{k}, \hat{i}+2\hat{j}-\hat{k}$ and $m\hat{i}-\hat{j}+2\hat{k}$ coplanar?
(a) 0 (b) 5/3
(c) 1 (d) 8/5
[NDA (II) - 2011]

14. If $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{a}.\vec{b}=0$ and $\vec{a}\times\vec{b}=\vec{0}$, then which one of the following is correct?
(a) $\vec{a}$ is parallel to $\vec{b}$
(b) $\vec{a}$ is perpendicular to $\vec{b}$
(c) either $\vec{a}$ or $\vec{b}$ is a null vector
(d) none of the above
[NDA-2011(2)]

15. The vectors $\hat{i}-x\hat{j}-y\hat{k}$ and $\hat{i}+x\hat{j}+y\hat{k}$ are orthogonal to each other, then what is the locus of the point (x, y)?
(a) A parabola (b) An ellipse
(c) A circle (d) A straight line
[NDA (I) - 2012]

16. If $\vec{a}$ = (2,1–1), $\vec{b}$ = (1,–1,0),c = (5,–1,1), then what is the unit vector parallel to $\vec{a}+\vec{b}-\vec{c}$ in the opposite direction?
(a) $\frac{\hat{i}+\hat{j}-2\hat{k}}{3}$ (b) $\frac{\hat{i}-2\hat{j}+2\hat{k}}{3}$

(c) $\frac{2\hat{i}-\hat{j}+2\hat{k}}{3}$ (d) None of these

[NDA (I) - 2012]

17. If the magnitudes of two vectors $\vec{a}$ and $\vec{b}$ are equal, then which one of the following is correct?

(a) $(\vec{a}+\vec{b})$ is parallel to $(\vec{a}-\vec{b})$

(b) $(\vec{a}+\vec{b}).(\vec{a}-\vec{b})=1$

(c) $(\vec{a}+\vec{b})$ is perpendicular to $(\vec{a}-\vec{b})$

(d) None of the above

[NDA (I) - 2012]

18. What is the value of m, if the vectors $2\hat{i}-\hat{j}+\hat{k}, \hat{i}+2\hat{j}-3\hat{k}$ and $3\hat{i}+m\hat{j}+5\hat{k}$ are coplanar?

(a) –2 (b) 2
(c) –4 (d) 4

[NDA (I) - 2012]

19. If $|\vec{a}|=10, |\vec{b}|=2$ and $\vec{a}.\vec{b}=12$, then what is the value of $|\vec{a}\times\vec{b}|$?

(a) 12 (b) 16
(c) 20 (d) 24

[NDA (I) - 2012]

20. What is the area of the rectangle having vertices A, B, C and D with position vectors $-\hat{i}+\frac{1}{2}\hat{j}+4\hat{k}, \hat{i}+\frac{1}{2}\hat{j}+4\hat{k}, \hat{i}-\frac{1}{2}\hat{j}+4\hat{k}, -\hat{i}-\frac{1}{2}\hat{j}+4\hat{k}$?

(a) 1/2sq unit (b) 1sq unit
(c) 2sq unit (d) 4sq units

[NDA (I) - 2012]

21. Let O be the origin and P, Q, R be the points such that $\overline{PO}+\overline{OQ}=\overline{QO}+\overline{OR}$. Then which one of the following is correct?

(a)P, Q, R are the vertices of an equilateral triangle
(b)P, Q, R are the vertices of an equilateral triangle
(c)P, Q, R are collinear
(d) None of above

[NDA (I) - 2012]

22. What is the value of λ for which $(\lambda\hat{i}+\hat{j}-\hat{k})\times(3\hat{i}-2\hat{j}+4\hat{k})=(2\hat{i}-11\hat{j}-7\hat{k})$?

(a) 2 (b) –2
(c) 1 (d) 7

[NDA (II) - 2012]

23. The vector $2\hat{j}-\hat{k}$ lies

(a) In the plane of XY (b) In the plane of YZ
(c) In the plane of XZ (d) Along the X-axis

[NDA (II) - 2012]

24. The vector $\vec{a}\times(\vec{b}\times\vec{a})$ is coplanar with:

(a) Only $\vec{a}$ (b) Only $\vec{b}$
(c) Both $\vec{a}$ and $\vec{b}$ a (c) Neither $\vec{a}$ nor $\vec{b}$

[NDA (II) - 2012]

25. EFGH is a rhombus such that the angle EFG is 60^0. The magnitude of vectors $\overline{FH}$ and $\{m\overline{EG}\}$ are equal where m is a scalar. What is the value of m?

(a) 3 (b) 1.5
(c) $\sqrt{2}$ (d) $\sqrt{3}$

[NDA (II) - 2012]

26. Consider the following

I. $4\hat{i}\times3\hat{i}=0$

II. $\frac{4\hat{i}}{3\hat{i}}=\frac{4}{3}$

Which of the above statements is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (II) - 2012]

27. Let ABCD is a parallelogram. If AB = $\vec{a}$ and BC = $\vec{b}$ then what is BD equal to?

(a) $\vec{a}+\vec{b}$ (b) $\vec{a}-\vec{b}$
(c) $-\vec{a}-\vec{b}$ (d) $-\vec{a}+\vec{b}$

[NDA (II) - 2012]

28. If $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{a}.\vec{b}=0$ and $\vec{a}\times\vec{b}=0$, then which one of the following is correct?

(a) $\vec{a}$ is parallel to $\vec{b}$

(b) $\vec{a}$ is perpendicular to $\vec{b}$

(c) Either $\vec{a}$ or $\vec{b}$ is a null vector

(d) None of the above

[NDA (II) - 2012]

29. The magnitude of the scalar p for which the vector p $(-3\hat{i}-2\hat{j}+13\hat{k})$ is of unit length is:

(a) 1/8 (b) 1/64
(c) $\sqrt{182}$ (d) $1/\sqrt{182}$

[NDA (II) - 2012]

30. For any vector α what $(\alpha.\hat{i})\hat{i}+(\alpha.\hat{j})\hat{j}+(\alpha.\hat{k})\hat{k}$ equal to?

(a) α (b) 3α
(c) –α (d) 0

[NDA (I) - 2013]

31. Which one of the following vectors is normal to the vector $\hat{i}+\hat{j}+\hat{k}$?

(a) $\hat{i}+\hat{j}-\hat{k}$ (b) $\hat{i}-\hat{j}+\hat{k}$
(c) $\hat{i}-\hat{j}-\hat{k}$ (d) None of these

[NDA (I) - 2013]

32. If θ is the angle between the vectors 4 $(\hat{i}-\hat{k})$ and $\hat{i}+\hat{j}+\hat{k}$, then what is (sinθ + cosθ) equal to?

(a) 0 (b) 1/2
(c) 1 (d) 2

[NDA (I) - 2013]

33. If the magnitude of $\vec{a}\times\vec{b}$ equals to $\vec{a}.\vec{b}$ then which one of the following is correct?

(a) $\vec{a}=\vec{b}$

(b) The angle between $\vec{a}$ and $\vec{b}$ a is 45°

(c) $\vec{a}$ is parallel to $\vec{b}$

(d) $\vec{a}$ is perpendicular to $\vec{b}$

[NDA (I) - 2013]

34. If $|\vec{a}|=\sqrt{2}, |\vec{b}|=\sqrt{3}$, and $|\vec{a}+\vec{b}|=\sqrt{6}$, then what is $|\vec{a}-\vec{b}|$ equal to?

(a) 1 (b) 2
(c) 3 (d) 4

[NDA (I) - 2013]

35. If $\vec{\beta}$ is perpendicular to both $\vec{\alpha}$ and $\vec{\gamma}$ where $\vec{\alpha} = \hat{k}$ and $\vec{\gamma} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, then what is $\vec{\beta}$ equal to?

(a) $3\hat{i} + 2\hat{j}$ (b) $-3\hat{i} + 2\hat{j}$

(c) $2\hat{i} - 3\hat{j}$ (d) $-2\hat{i} + 3\hat{j}$

[NDA (I) - 2013]

36. What is the area of the triangle whose vertices are (0,0,0), (1,2,3)and (–3,–2,1)?

(a) $3\sqrt{5}$ (b) $6\sqrt{5}$

(c) 6 square unit (d) 12 square unit

[NDA-2013(1)]

37. If the angle between the vectors $\hat{i} - m\hat{j}$ and $\hat{j} + \hat{k}$ is $\frac{\pi}{3}$, then what is the value of m?

(a) 0 (b) 2

(c) –2 (d) None of these

[NDA (II) - 2013]

38. The vectors $\hat{i} - 2x\hat{j} - 3y\hat{k}$ and $\hat{i} + 3x\hat{j} + 2y\hat{k}$ are orthogonal to each other. Then, the locus of the point (x, y) is:

(a) Hyperbola (b) Ellipse

(c) Parabola (d) Circle

[NDA (II) - 2013]

39. What is the value of p for which the vector $\vec{p}(2\hat{i} - \hat{j} + 2\hat{k})$ is of 3 unit's length?

(a) 1 (b) 2

(c) 3 (d) 6

[NDA (II) - 2013]

40. If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$ $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$ $\vec{c} = 3\hat{i} + \hat{j}$ are three vectors such that $\vec{a} + t\vec{b}$ is perpendicular to c, then what is t equal to?

(a) 8 (b) 6

(c) 4 (d) 2

[NDA (II) - 2013]

41. What is the vector perpendicular to both the vector $\hat{i} - \hat{j}$ and $\hat{i}$?

(a) $\hat{i}$ (b) $-\hat{j}$

(c) $\hat{j}$ (d) $\hat{k}$

[NDA (II) - 2013]

42. The position vectors of the points A and B are respectively $3\hat{i} - 5\hat{j} + 2\hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$. What is the length of AB?

(a) 11 (b) 9

(c) 7 (d) 0

[NDA (II) - 2013]

Directions (for next three): The vertices of a ΔABC are A (2, 3, 1), B (–2, 2, 0) and C (0,1,–1).

43. What is the cosine of angle ABC?

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{2}}$

(c) $\frac{2}{\sqrt{6}}$ (d) None of these

[NDA (I) - 2014]

44. What is the area of the triangle?

(a) $6\sqrt{6}$ sq units (b) $3\sqrt{2}$ sq units

(c) $10\sqrt{3}$ sq units (d) None of these

[NDA (I) - 2014]

45. What is the magnitude of the line joining mid-points of the sides AC and BC?

(a) $\frac{1}{\sqrt{2}}$ unit (b) 1 unit

(c) $\frac{3}{\sqrt{2}}$ units (d) 2 units

[NDA (I) - 2014]

Directions (for next two): Let $|\vec{a}| = 7$, $|\vec{b}| = 11$, $|\vec{a} + \vec{b}| = 10\sqrt{3}$

46. What is $|\vec{a} - \vec{b}|$ equal to?

(a) $2\sqrt{2}$ (b) $2\sqrt{10}$

(c) 5 (d) 10

[NDA (I) - 2014]

47. What is the angle between $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$?

(a) π/2 (b) π/3

(c) π/6 (d) None of these

[NDA (I) - 2014]

Directions (for next two): Consider the vectors $\vec{a}$ = i – 2j + k and $\vec{b}$ = 4i – 4j + 7k.

48. What is the scalar projection of $\vec{a}$ on $\vec{b}$?

(a) 1 (b) $\frac{19}{9}$

(c) $\frac{17}{9}$ (d) $\frac{23}{9}$

[NDA (I) - 2014]

49. What is the vector perpendicular to both the vectors?

(a) –10i – 3j + 4k (b) –10i + 3j + 4k

(c) 10i – 3j + 4k (d) None of these

[NDA (I) - 2014]

50. If $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, then what is a. b equal to?

(a) 6 (b) 7

(c) 8 (d) 9

[NDA (II) - 2014]

51. If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then which one of the following is correct?

(a) $|\vec{a}| = |\vec{b}|$

(b) a is parallel to b

(c) a is perpendicular to b

(d) $\hat{a}$ is a unit vector

[NDA (II) - 2014]

52. What is the area of ΔOAB, where O is the origin, $\overrightarrow{OA} = 3\hat{i} - \hat{j} + \hat{k}$ and $\overrightarrow{OB} = 2\hat{i} + \hat{j} - 3\hat{k}$?

(a) $5\sqrt{6}$ sq units (b) $\frac{5\sqrt{6}}{2}$ sq units

(c) $\sqrt{6}$ sq units (d) $\sqrt{30}$ sq units

[NDA (II) - 2014]

53. Which one of the following is the unit vector perpendicular to both $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$?

(a) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (b) $\hat{k}$

(c) $\frac{\hat{j} + \hat{k}}{\sqrt{2}}$ (d) $\frac{\hat{i} - \hat{j}}{\sqrt{2}}$

[NDA (II) - 2014]

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54. What is the interior acute angle of the parallelogram whose sides are represented by the vectors $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i}-\frac{1}{\sqrt{2}}\hat{j}+\hat{k}$?

(a) 60° (b) 45°
(c) 30° (d) 15°

[NDA (II) - 2014]

55. For what value of λ are the vectors $\lambda\hat{i}+(1+\lambda)\hat{j}+(1+2\lambda)\hat{k}$ and $(1-\lambda)\hat{i}+\lambda\hat{j}+2\hat{k}$ perpendicular?

(a) –1/3 (b) 1/3
(c) 2/3 (d) 1

[NDA (II) - 2014]

Directions (for next four): Read the following information carefully and answer the questions given below.

$\vec{a}+\vec{b}+\vec{c}=0$ such that $|\vec{a}|=3$, $|\vec{b}|=5$ and $|\vec{c}|=7$

56. What is the angle between $\vec{a}$ and $\vec{b}$?

(a) π/6 (b) π/4
(c) π/3 (d) π/2

[NDA (II) - 2014]

57. What is $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$ equal to?

(a) –83 (b) $-\frac{83}{2}$
(c) 75 (d) $-\frac{75}{2}$

[NDA (II) - 2014]

58. What is cosine of the angle between $\vec{b}$ and $\vec{c}$?

(a) $\frac{11}{12}$ (b) $\frac{13}{14}$
(c) $-\frac{11}{12}$ (d) $-\frac{13}{14}$

[NDA (II) - 2014]

59. What is $|\vec{a}+\vec{b}|$ equal to?

(a) 7 (b) 8
(c) 10 (d) 11

[NDA (II) - 2014]

60. The adjacent sides AB and AC of a triangle ABC are represented by the vectors $-2\hat{i}+3\hat{j}+2\hat{k}$ and $-4\hat{i}+5\hat{j}+2\hat{k}$ respectively. The area of the triangle ABC is:

(a) 6 sq units (b) 5 sq units
(c) 4 sq units (d) 3 sq units

[NDA (I) - 2015]

61. A force F = $3\hat{i}+4\hat{j}-3\hat{k}$ is applied at the point P, whose position vector is r = $2\hat{i}-2\hat{j}-3\hat{k}$. What is the magnitude of the moment of the force about the origin?

(a) 23 units (b) 19 units
(c) 18 units (d) 21 units

[NDA (I) - 2015]

62. If $|\vec{a}|=7$, $|\vec{b}|=11$ and $|\vec{a}+\vec{b}|=10\sqrt{3}$, then $|\vec{a}-\vec{b}|$ is equal to:

(a) 40 (b) 10
(c) $4\sqrt{10}$ (d) $2\sqrt{10}$

[NDA (I) - 2015]

63. Let α, β and γ be distinct real numbers. The points with position vectors $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$, $\beta\hat{i}+\gamma\hat{j}+\alpha\hat{k}$ and $\gamma\hat{i}+\alpha\hat{j}+\beta\hat{k}$

(a) Are collinear
(b) Form an equilateral triangle
(c) Form a scalene triangle
(d) Form a right-angled triangle

[NDA (I) - 2015]

64. If $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$, then which one of the following is correct?

(a) $\vec{a}=\lambda\vec{b}$ for some scalar λ
(b) $\vec{a}$ is parallel to $\vec{b}$
(c) $\vec{a}$ is perpendicular to $\vec{b}$
(d) $\vec{a}=\vec{b}=0$

[NDA (I) - 2015]

65. Given that the vector α and β are non-collinear. The values of x and y for which u–v = w holds true if u = 2xα + yβ, v = 2yα + 3xβ and w = 2α – 5β, are

(a) x=2, y=1 (b) x=1, y=2
(c) x= – 2, y = 1 (d) x = –2, y = –1

[NDA (I) - 2015]

66. If $\vec{a}+\vec{b}+\vec{c}=0$, then which of the following is/are correct?

1. $\vec{a}$, $\vec{b}$, $\vec{c}$ are coplanar
2. $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$

Select the correct answer using the codes given below:

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2015]

67. If the vectors $\alpha\hat{i}+\alpha\hat{j}+\gamma\hat{k}$, $\hat{i}+\hat{k}$ and $\gamma\hat{i}+\gamma\hat{j}+\beta\hat{k}$ lie on a plane, where α, β and γ are distinct non-negative numbers, they γ is:

(a) Arithmetic mean of α and β
(b) Geometric mean of α and β
(c) Harmonic mean of α and β
(d) None of the above

[NDA (II) - 2015]

68. The area of the square, one of whose diagonals is $3\hat{i}+4\hat{j}$ is:

(a) 12 square unit (b) 12.5 square unit
(c) 25 square unit (d) 156.25square unit

[NDA (II) - 2015]

69. ABCD is a parallelogram and P is the point of intersection of the diagonals. If O is the origin, then $\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}$ is equal to:

(a) $4\overrightarrow{OP}$ (b) $2\overrightarrow{OP}$
(c) $\overrightarrow{OP}$ (d) Null vector

[NDA (II) - 2015]

70. If $\vec{b}$ and $\vec{c}$ are the position vectors of the points B and C respectively, then the position vector of the point D such that $\overrightarrow{BD}=4\overrightarrow{BC}$ is:

(a) $4(\vec{c}-\vec{b})$ (b) $-4(\vec{c}-\vec{b})$
(c) $4\vec{c}-3\vec{b}$ (d) $4\vec{c}+3\vec{b}$

[NDA (II) - 2015]

71. If the position vector $\vec{a}$ of the point (5, n) is such that $|\vec{a}|=13$, then the value/values of n can be.

(a) ±8 (b) ±12
(c) Only 8 (d) Only 12

[NDA (II) - 2015]

72. If $|\vec{a}|=2$ and $|\vec{b}|=3$, then $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2$ is equal to:

(a) 72 (b) 64
(c) 48 (d) 36

[NDA (II) - 2015]

73. Consider the following inequalities in respect of vector $\vec{a}$ and $\vec{b}$.

1. $|\vec{a}+\vec{b}| \le |\vec{a}| + |\vec{b}|$

2. $|\vec{a}-\vec{b}| \le |\vec{a}| - |\vec{b}|$

Which of the above is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2015]

74. If the magnitude of difference of two unit vectors is $\sqrt{3}$, then the magnitude of sum of the two vectors is:
(a) 1/2 unit (b) 1 unit
(c) 2 unit (d) 3 unit

[NDA (II) - 2015]

75. The vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are such that $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$. Which of the following is/are correct?

1. $(\vec{a}-\vec{d})\times(\vec{b}-\vec{c}) = \vec{0}$

2. $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = \vec{0}$

Select the correct answer using the codes given below:
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2015]

76. What is vector of unit length orthogonal to both the vectors $\hat{i}+\hat{j}+\hat{k}$ and $2\hat{i}+3\hat{j}-\hat{k}$?

(a) $\frac{-4\hat{i}+3\hat{j}-\hat{k}}{\sqrt{26}}$ (b) $\frac{-4\hat{i}+3\hat{j}+\hat{k}}{\sqrt{26}}$

(c) $\frac{-3\hat{i}+2\hat{j}-\hat{k}}{\sqrt{14}}$ (d) $\frac{-3\hat{i}+2\hat{j}+\hat{k}}{\sqrt{14}}$

[NDA (I) - 2016]

77. If $\vec{a}, \vec{b}$ and $\vec{c}$ are the position vectors of the vertices of an equilateral triangle whose orthocenter is at the origin, and then which one of the following is correct?

(a) $\vec{a}+\vec{b}+\vec{c} = \vec{0}$

(b) $\vec{a}+\vec{b}+\vec{c}$ =unit vector

(c) $\vec{a}+\vec{b} = \vec{c}$

(d) $\vec{a} = \vec{b}+\vec{c}$

[NDA (I) - 2016]

78. What is the area of the parallelogram having diagonals $3\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}-3\hat{j}+4\hat{k}$?

(a) $5\sqrt{3}$ square units (b) $4\sqrt{5}$ square units

(c) $3\sqrt{3}$ square units (d) $15\sqrt{2}$ square units

[NDA (I) - 2016]

For the next two (2) items that follow:

Let $\hat{a}, \hat{b}$ be two unit vectors and θ be the angle between them.

79. What is cos $\left(\frac{\theta}{2}\right)$ equal to?

(a) $\frac{|\hat{a}-\hat{b}|}{2}$ (b) $\frac{|\hat{a}+\hat{b}|}{2}$

(c) $\frac{|\hat{a}-\hat{b}|}{4}$ (d) $\frac{|\hat{a}+\hat{b}|}{4}$

[NDA (I) - 2016]

80. What is sin $\left(\frac{\theta}{2}\right)$ equal to:

(a) $\frac{|\hat{a}-\hat{b}|}{2}$ (b) $\frac{|\hat{a}+\hat{b}|}{2}$

(c) $\frac{|\hat{a}-\hat{b}|}{4}$ (d) $\frac{|\hat{a}+\hat{b}|}{4}$

[NDA (I) - 2016]

Direction (for next two): Consider the following for the next two items that follow:

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a}+\vec{b}+\vec{c} = \vec{0}$, and $|\vec{a}| = 10, |\vec{b}| = 6$ and $|\vec{c}| = 14$.

81. What is $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ equal to?
(a) –332 (b) –166
(c) 0 (d) 166

[NDA (II) - 2016]

82. What is the angle between $\vec{a}$ and $\vec{b}$?
(a) 30° (b) 45°
(c) 60° (d) 75°

[NDA (II) - 2016]

83. In right-angled triangle ABC, if the hypotenuse AB = p, then what is: $\overrightarrow{AB}.\overrightarrow{AC}+\overrightarrow{BC}.\overrightarrow{BA}+\overrightarrow{CA}.\overrightarrow{CB}$ equal to?
(a) p (b) p^2

(c) $2p^2$ (d) $\frac{p^2}{2}$

[NDA (II) - 2016]

84. A force $\vec{F} = 3i+2j-4\hat{k}$ is applied at the point (1, –1, 2). What is the moment of the force about the point (2,–1,3)?

(a) $\hat{i}+4\hat{j}+4\hat{k}$ (b) $2\hat{i}+\hat{j}+2\hat{k}$

(c) $2\hat{i}-7\hat{j}-2\hat{k}$ (d) $2\hat{i}+4\hat{j}+\hat{k}$

[NDA (II) - 2016]

Direction (for next two): Consider the following for the next two items that follow:

Let $\vec{a} = \hat{i}+\hat{j}, \vec{b} = 3\hat{i}+4\hat{k}$ and $\vec{b} = \vec{c}+\vec{d}$, where $\vec{c}$ is parallel to $\vec{a}$ and $\vec{d}$ is perpendicular to $\vec{a}$.

85. What is $\vec{c}$ equal to?

(a) $\frac{3(\hat{i}+\hat{j})}{2}$ (b) $\frac{2(\hat{i}+\hat{j})}{3}$

(c) $\frac{(\hat{i}+\hat{j})}{2}$ (d) $\frac{(\hat{i}+\hat{j})}{3}$

[NDA (II) - 2016]

86. If $\vec{d} = xi+yj+z\hat{k}$, then which of the following equations is/are correct?
1. y–x = 4
2. 2z–3=0
Select the correct answer using the code given below:
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2016]

87. If $\vec{a} = \hat{i}-\hat{j}+\hat{k}, \vec{b} = 2\hat{i}+3\hat{j}+2\hat{k}$ and $\vec{c} = \hat{i}+m\hat{j}+n\hat{k}$ are three coplanar vectors and $|\vec{c}| = \sqrt{6}$, then which one of the following is correct?
(a) m = 2 and n = ±1 (b) m = ± 2 and n = –1
(c) m = 2 and n = –1 (d) m = ± 2 and n =1

[NDA (I) - 2017]

88. If $\vec a\times\vec b=\vec c$ and $\vec b\times\vec c=\vec a$, then which one of the following is correct?

(a) $\vec a, \vec b, \vec c$ are orthogonal in pairs and $|\vec a|=|\vec c|$ and $|\vec b|=1$

(b) $\vec a, \vec b, \vec c$ are non-orthogonal to each other

(c) $\vec a, \vec b, \vec c$ are orthogonal in pairs but $|\vec a|\neq|\vec c|$

(d) $\vec a, \vec b, \vec c$ are orthogonal in pairs but $|\vec b|\neq 1$

[NDA (I) - 2017]

89. Let ABCD be a parallelogram whose diagonals intersect at P and let O be the origin. What is $\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}$ equal to:

(a) $2\overrightarrow{OP}$ (b) $4\overrightarrow{OP}$
(c) $6\overrightarrow{OP}$ (d) $8\overrightarrow{OP}$

[NDA (I) - 2017]

90. ABCD is a quadrilateral whose diagonals are AC and BD. Which one of the following is correct?

(a) $\overrightarrow{BA}+\overrightarrow{CD}=\overrightarrow{AC}+\overrightarrow{DB}$ (b) $\overrightarrow{BA}+\overrightarrow{CD}=\overrightarrow{BD}+\overrightarrow{CA}$
(c) $\overrightarrow{BA}+\overrightarrow{CD}=\overrightarrow{AC}+\overrightarrow{BD}$ (d) $\overrightarrow{BA}+\overrightarrow{CD}=\overrightarrow{BC}+\overrightarrow{AD}$

[NDA (I) - 2017]

91. If $\vec a = 2\hat i+3\hat j+4\hat k$ and $\vec b = 3\hat i+2\hat j-\lambda\hat k$ are perpendi- cular, then what is the value of λ?

(a) 2 (b) 3
(c) 4 (d) 5

[NDA (I) - 2017]

92. If α, β and γ are angles which he vector $\overrightarrow{OP}$ (O being the origin) makes with positive direction of the coordinate axes, then which of the following are correct?

1. $\cos^2\alpha + \cos^2\beta = \sin^2\gamma$
2. $\sin^2\alpha + \cos^2\beta = \cos^2\gamma$
3. $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Select the correct answer using the codes given below:

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

[NDA (II) - 2017]

93. Let $\vec\alpha = \hat i+2\hat j-\hat k$, $\vec\beta = 2\hat i-\hat j+3\hat k$ and $\vec\gamma = 2\hat i+\hat j+6\hat k$ be three vector. If $\vec\alpha$ and $\vec\beta$ are both perpendicular to the vector $\vec\delta$ and $\vec\delta.\vec\gamma=10$, then what is the magnitude of $\vec\delta$?

(a) $\sqrt3$ units (b) $2\sqrt3$ units
(c) $\frac{\sqrt3}{2}$ units (d) $\frac{1}{\sqrt3}$ units

[NDA (II) - 2017]

94. If $\hat a$ and $\hat b$ are two unit vectors, then the vector $(\hat a+\hat b)\times(\hat a\times\hat b)$ is parallel to:

(a) $(\hat a-\hat b)$ (b) $(\hat a+\hat b)$
(c) $(2\hat a-\hat b)$ (d) $(2\hat a+\hat b)$

[NDA (II) - 2017]

95. A force $\vec F = \hat i+3\hat j+2\hat k$ acts on a particle to displace it from the point A $(\hat i+2\hat j-3\hat k)$ to the point B $(3\hat i-\hat j+5\hat k)$. The work done by the force will be:

(a) 6 units (b) 7 units
(c) 9 units (d) 10 units

[NDA (II) - 2017]

96. For any vector $\vec a$ $|\vec a\times\hat i|^2+|\vec a\times\hat j|^2+|\vec a\times\hat k|^2$ is equal to:

(a) $|\vec a|^2$ (b) $2|\vec a|^2$
(c) $3|\vec a|^2$ (d) $4|\vec a|^2$

[NDA (II) - 2017]

97. If the vectors $a\hat i+\hat j+\hat k, \hat i+b\hat j+\hat k$ and $\hat i+\hat j+c\hat k$ (a, b, c $\neq 1$) are coplanar, then the value of $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$ is equal to:

(a) 0 (b) 1
(c) a + b + c (d) abc

[NDA (II) - 2017]

98. If the vectors $\vec k$ and $\vec A$ are parallel to each other, then what is $k\vec k\times\vec A$ equal to?

(a) $k^2\ \vec A$ (b) $\vec 0$
(c) $-k^2\ \vec A$ (d) $\vec A$

[NDA (I) - 2018]

99. If $\vec a+2\vec b+3\vec c=\vec 0$ and $\vec a\times\vec b+\vec b\times\vec c+\vec c\times\vec a=\lambda(\vec b\times\vec c)$ then what is the value of λ?

(a) 2 (b) 3
(c) 4 (d) 6

[NDA (I) - 2018]

100. What is the moment about the point $\hat i+2\hat j-\hat k$ of a force represented by $3\hat i+\hat k$ acting through the point $2\hat i-\hat j+3\hat k$?

(a) $-3\hat i+11\hat j+9\hat k$ (b) $3\hat i+2\hat j+9\hat k$
(c) $3\hat i+4\hat j+9\hat k$ (d) $\hat i+\hat j+\hat k$

[NDA (I) - 2018]

101. Let $\vec p$ and $\vec q$ be the position vectors of the point P and Q respectively with respect to origin O. The points R and S divide PQ internally and externally respectively in the ratio 2:3. If $\overrightarrow{OR}$ and $\overrightarrow{OS}$ are perpendicular, then which one of the following is correct?

(a) $9p^2 = 4q^2$ (b) $4p^2 = 9q^2$
(c) $9p = 4q$ (d) $4p = 9q$

[NDA (I) - 2018]

102. If $\vec a$ and $\vec b$ are vectors such that $|\vec a|=2, |\vec b|=7$ and $\vec a\times\vec b=3\hat i+2\hat j+6\hat k$, then what is the acute angle between $\vec a$ and $\vec b$?

(a) 30° (b) 45°
(c) 60° (d) 90°

[NDA (I) - 2018]

103. In a triangle ABC, if taken in order, consider the following statements:

1. $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\vec 0$
2. $\overrightarrow{AB}+\overrightarrow{BC}-\overrightarrow{CA}=\vec 0$
3. $\overrightarrow{AB}-\overrightarrow{BC}+\overrightarrow{CA}=\vec 0$
4. $\overrightarrow{BA}-\overrightarrow{BC}+\overrightarrow{CA}=\vec 0$

How many of the above statements are correct?

(a) One (b) Two
(c) Three (d) Four

[NDA (II) - 2018]

104. A spacecraft at $\hat i+2\hat j+3\hat k$ is subjected to a force $\lambda\hat k$ by firing a rocket. The spacecraft is subjected to a moment of magnitude.

(a) λ (b) $\sqrt3\lambda$
(c) $\sqrt5\lambda$ (d) None of these

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[NDA (II) - 2018]

105. What is $(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})$ equal to?

(a) $\vec{0}$ (b) $\vec{a}\times\vec{b}$

(c) $2(\vec{a}\times\vec{b})$ (d) $|\vec{a}|^2-|\vec{b}|^2$

[NDA (II) - 2018]

106. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three mutually perpendicular vectors each of unit magnitude. If $\vec{A}=\vec{a}+\vec{b}+\vec{c}$, $\vec{B}=\vec{a}-\vec{b}+\vec{c}$ and $\vec{C}=\vec{a}-\vec{b}-\vec{c}$, then which one of the following is correct?

(a) $|\vec{A}|>|\vec{B}|>|\vec{C}|$ (b) $|\vec{A}|=|\vec{B}|\neq|\vec{C}|$

(c) $|\vec{A}|=|\vec{B}|=|\vec{C}|$ (d) $|\vec{A}|\neq|\vec{B}|\neq|\vec{C}|$

[NDA (II) - 2018]

107. If $|\vec{a}|=3, |\vec{b}|=4$ and $|\vec{a}-\vec{b}|=5$, then what is the value of $|\vec{a}+\vec{b}|=?$

(a) 8 (b) 6

(c) $5\sqrt{2}$ (d) 5

[NDA (II) - 2018]

108. A unit vector perpendicular to each of the vectors $2\hat{i}-\hat{j}+\hat{k}$ and $3\hat{i}-4\hat{i}-\hat{k}$ is:

(a) $\frac{1}{\sqrt{3}}\hat{i}+\frac{1}{\sqrt{3}}\hat{j}-\frac{1}{\sqrt{3}}\hat{k}$ (b) $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{2}\hat{j}+\frac{1}{2}\hat{k}$

(c) $\frac{1}{\sqrt{3}}\hat{i}-\frac{1}{\sqrt{3}}\hat{j}-\frac{1}{\sqrt{3}}\hat{k}$ (d) $\frac{1}{\sqrt{3}}\hat{i}+\frac{1}{\sqrt{3}}\hat{j}+\frac{1}{\sqrt{3}}\hat{k}$

[NDA (II) - 2018]

109. If $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$, then what is $\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})$ equal to?

(a) x (b) x + y

(c) –(x + y + z) (d) (x + y + z)

[NDA (II) - 2018]

110. Let $|\vec{a}|\neq 0, |\vec{b}|\neq 0$ $(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})=|\vec{a}|^2+|\vec{b}|^2$ holds if and only if

(a) $\vec{a}$ and $\vec{b}$ are perpendicular

(b) $\vec{a}$ and $\vec{b}$ are parallel

(c) $\vec{a}$ and $\vec{b}$ are inclined at an angle of 45°

[NDA (II) - 2018]

111. If $\vec{a}=\hat{i}-2\hat{j}+5\hat{k}$, and $\vec{b}=2\hat{i}+\hat{j}-3\hat{k}$, then $(\vec{b}-\vec{a})\cdot(3\vec{a}+\vec{b})$ equal to?

(a) 106 (b) −106

(c) 53 (d) −53

[NDA (I) - 2019]

112. If position vectors of two points A and B are $3\hat{i}-2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}-3\hat{k}$, respectively. Then length of $\overline{AB}$ is equal to?

(a) $\sqrt{14}$ (b) $\sqrt{29}$

(c) $\sqrt{43}$ (d) $\sqrt{53}$

[NDA (I) - 2019]

113. In a right angled triangle hypotenuse AC = p, then $\overline{AB}\cdot\overline{AC}+\overline{BC}\cdot\overline{BA}+\overline{CA}\cdot\overline{CB}$ equal to ?

(a) p^2 (b) $2p^2$

(c) $\frac{p^2}{2}$ (d) p

[NDA (I) - 2019]

114. Find the sine of angles between vectors $\vec{a}=2\hat{i}-6\hat{j}-3\hat{k}$, and $\vec{b}=4\hat{i}+3\hat{j}-\hat{k}$?

(a) $\frac{1}{\sqrt{26}}$ (b) $\frac{5}{\sqrt{26}}$

(c) $\frac{5}{26}$ (d) $\frac{1}{26}$

[NDA (I) - 2019]

115. Find the value of λ for which $3\hat{i}+4\hat{j}-\hat{k}$, and $-2\hat{i}+\lambda\hat{j}+10\hat{k}$ are perpendicular?

(a) 1 (b) 2

(c) 3 (d) 4

[NDA (I) - 2019]

116. What is the scalar projection of $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$, on $\vec{b}=4\hat{i}-4\hat{j}+7\hat{k}$?

(a) $\frac{\sqrt{9}}{6}$ (b) $\frac{19}{9}$

(c) $\frac{9}{19}$ (d) $\frac{\sqrt{6}}{19}$

[NDA (II) - 2019]

117. If the magnitude of the sum of two non-zero vectors is equal to magnitude of their difference, then which one of the following is correct?

(a) The vectors are parallel

(b) The vectors are perpendicular

(c) The vectors are anti-parallel

(d) The vectors must be unit vectors

[NDA (II) - 2019]

118. Consider the following equations for two vectors $\vec{a}$ and $\vec{b}$:

1. $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=|\vec{a}|^2-|\vec{b}|^2$

2. $(|\vec{a}+\vec{b}|)\cdot(|\vec{a}-\vec{b}|)=|\vec{a}|^2-|\vec{b}|^2$

3. $|\vec{a}\cdot\vec{b}|^2+|\vec{a}\times\vec{b}|^2=|\vec{a}|^2|\vec{b}|^2$

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only

(c) 1 and 3 only (d) 2 and 3 only

[NDA (II) - 2019]

119. Consider the following statements:

1.The magnitude of $\vec{a}\times\vec{b}$ is same as the area of a triangle with sides $\vec{a}$ and $\vec{b}$

2.If $\vec{a}\times\vec{b}=\vec{0}$, where $\vec{a}\neq\vec{0}$, $\vec{b}\neq 0$ then $\vec{a}=\lambda\vec{b}$

which of the following statements is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2019]

120. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then what is $\sin^2\left(\frac{\theta}{2}\right)$ equal to ?

(a) $\frac{|\vec{a}+\vec{b}|^2}{4}$ (b) $\frac{|\vec{a}-\vec{b}|^2}{4}$

(c) $\frac{|\vec{a}+\vec{b}|^2}{2}$ (d) $\frac{|\vec{a}-\vec{b}|^2}{2}$

[NDA (II) - 2019]

121. If $\hat{a}$ is a unit vector in the XY plane making an angle 30° with the positive x-axis, then what is the $\hat{a}$ equal to ?

(a) $\frac{\sqrt{3}\hat{i}+\hat{j}}{2}$ (b) $\frac{\sqrt{3}\hat{i}-\hat{j}}{2}$

(c) $\frac{\hat{i}+\sqrt{3}\hat{j}}{2}$ (d) $\frac{\hat{i}-\sqrt{3}\hat{j}}{2}$

[NDA 2020]

122. Let A be a point in space such that $|\overline{OA}|$ = 12, where O is the origin. If $\overline{OA}$ is inclined at 45° and 60° with x-axis and y-axis respectively. Then what is $\overline{OA}$ is equal to

(a) $6\hat{i}+6\hat{j}\pm\sqrt{2}\hat{k}$ (b) $6\hat{i}+6\sqrt{2}\hat{j}\pm 6\hat{k}$

(c) $6\sqrt{2}\hat{i}+6\hat{j}\pm 6\hat{k}$ (d) $3\sqrt{2}\hat{i}+3\hat{j}\pm 6\hat{k}$

[NDA 2020]

123. The adjacent sides of a parallelogram are $2\hat{i}-4\hat{j}+5\hat{k}$ and $\hat{i}-2\hat{j}-3\hat{k}$ .What is the magnitude of dot product of vectors which represents its diagonals?

(a) 21 (b) 25
(c) 31 (d) 36

[NDA 2020]

124. If $|\vec{a}\times\vec{b}|^2 + |\vec{a}.\vec{b}|^2 = 144$, and $|\vec{a}|$ = 4 then what is $|\vec{b}|$ is equal to:

(a) 3 (b) 4
(c) 6 (d) 8

[NDA 2020]

125. If the vectors $\vec{a} = 2\hat{i}-3\hat{j}+\hat{k}$, $\vec{b} = \hat{i}+2\hat{j}-3\hat{k}$, $\vec{c} = \hat{j}+p\hat{k}$, are coplanar, then what is the value of p?

(a) 1 (b) −1
(c) 5 (d) −5

[NDA 2020]

126. A vector $\vec{r} = a\hat{i}+b\hat{j}$ is equally inclined to both x and y axes. If the magnitude of the vector is 2 units, then what are the values of a and b respectively?

(a) $\frac{1}{2},\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}$

(c) $\sqrt{2},\sqrt{2}$ (d) 2, 2

[NDA (I) - 2021]

127. Consider the following statements in respect of a vector $\vec{c} = \vec{a}+\vec{b}$, where $|\vec{a}| = |\vec{b}| \neq 0$:

1. $\vec{c}$ is perpendicular to $(\vec{a}-\vec{b})$
2. $\vec{c}$ is perpendicular to $(\vec{a}\times\vec{b})$

Which of the above statements is/are correct?

(a) 1only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2021]

128. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}| = 4$, then which one of the following is correct?

(a) $\vec{a}$ and $\vec{b}$ must be unit vectors

(b) $\vec{a}$ must be parallel to $\vec{b}$

(c) $\vec{a}$ must be perpendicular to $\vec{b}$

(d) $\vec{a}$ must be equal to $\vec{b}$

[NDA (I) - 2021]

129. If $\vec{a},\vec{b}$ and $\vec{c}$ are coplanar, then what is $(2\vec{a}\times 3\vec{b}).4\vec{c}+(5\vec{b}\times 3\vec{c}).6\vec{a}$ equal to?

(a) 114 (b) 66
(c) 0 (d) –66

[NDA (I) - 2021]

130. Consider the following statements:

1.The cross product of two unit vectors is always a unit vector
2.The dot product of two unit vectors is always unit
3.The magnitude of sum of two unit vectors is always greater than the magnitude of their difference.

Which of the above statements are not correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (I) - 2021]

131. Let $\vec{a},\vec{b}$ and $\vec{c}$ be unit vectors such that $\vec{a}\times\vec{b}$ is perpendicular to $\vec{c}$. If θ is the angle between $\vec{a}$ and $\vec{b}$, then which of the following is/are correct?

1. $\vec{a}\times\vec{b} = \sin\theta\,\vec{c}$
2. $\vec{a}.(\vec{b}\times\vec{c})=0$

Select the correct answer using the code given below:

(a)1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2021]

132. If $\vec{a}+3\vec{b} = 3\hat{\imath}-\hat{\jmath}$ and $2\vec{a}+\vec{b} = \hat{\imath}-2\hat{\jmath}$, then what is the angle between $\vec{a}$ and $\vec{b}$?

(a) 0 (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

[NDA (II) - 2021]

133. If $(\vec{a}+\vec{b})$ is perpendicular to $\vec{a}$ and magnitude of $\vec{b}$ is twice that of $\vec{a}$, then what is the value of $(4\vec{a}+\vec{b}).\vec{b}$ equal to?

(a) 0 (b) 1
(c) $8|\vec{a}|^2$ (d) $8|\vec{b}|^2$

[NDA (II) - 2021]

134. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a},\vec{b}$ and $\vec{c}$ are coplanar. Which of the following is/are correct?

1. $(\vec{a}\times\vec{b})\times\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$
2. $(\vec{a}\times\vec{b})\times\vec{c}$ is perpendicular to $\vec{a}\times\vec{b}$

Select the correct answer using the code given below:

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2021]

135. If the position vectors of A and B are $(\sqrt{2}-1)\hat{i}-\hat{j}$ and $\hat{i}+(\sqrt{2}+1)\hat{j}$ respectively, then what is the magnitude of $\overrightarrow{AB}$?

(a) $2\sqrt{2}$ (b) $3\sqrt{2}$

(c) $2\sqrt{3}$ (d) $3\sqrt{3}$

[NDA (II) - 2021]

136. If $4\hat{\imath}+\hat{\jmath}-3\hat{k}$ and $p\hat{\imath}+q\hat{\jmath}-2\hat{k}$ are collinear vectors, then what are the possible values of p and q respectively?

(a) 4, 1 (b) 1, 4
(c) $\frac{8}{3},\frac{2}{3}$ (d) $\frac{2}{3},\frac{8}{3}$

[NDA (I) - 2022]

137. If $\vec{a}. \vec{b}, \vec{c}$ are the position vectors of the vectors A, B, C respectively of a triangle ABC and G is the centroid of the triangle, then what is $\overrightarrow{AG}$ equal to?

(a) $\frac{\vec{a}+\vec{b}+\vec{c}}{3}$ (b) $\frac{2\vec{a}-\vec{b}-\vec{c}}{3}$

(c) $\frac{\vec{b}+\vec{c}-2\vec{a}}{3}$ (d) $\frac{\vec{a}-2\vec{b}-2\vec{c}}{3}$

[NDA (I) - 2022]

138. Consider the followings:
1.Dot product over vector addition is distributive
2.Cross product over vector addition is distributive
3.Cross product of vectors is associative
Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

[NDA (I) - 2022]

139. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors such that $\vec{a} \times \vec{b} = \vec{c}$. Consider the following statement:
1. $\vec{a}$ is unique if $\vec{b}$ and $\vec{c}$ are given
2. $\vec{c}$ is unique if $\vec{a}$ and $\vec{b}$ are given
Which of the above statements is/are correct?

[NDA (I) - 2022]

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

140. Let $\vec{a}, and\ \vec{b}$ be two unit vectors such that $|\vec{a}-\vec{b}| < 2$. If 2θ is the angle between $\vec{a}\ and\ \vec{b}$, then which one of the following is correct?

[NDA (I) - 2022]

(a) $0 < \sin\theta < 1$ only (b) $-\frac{1}{2} < \sin\theta < \frac{1}{2}$ only
(c) $-1 < \sin\theta < 0$ only (d) $-1 < \sin\theta < 1$

141. PQRS is a parallelogram. If $\overrightarrow{PR} = \vec{a}$ and $\overrightarrow{QS} = \vec{b}$, then what is $\overrightarrow{PQ}$ equal to ?

(a) $\vec{a}+\vec{b}$ (b) $\vec{a}-\vec{b}$

(c) $\frac{\vec{a}+\vec{b}}{2}$ (d) $\frac{\vec{a}-\vec{b}}{2}$

[NDA 2022 (II)]

142. Let $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a}+2\vec{b}$ and $5\vec{a}-4\vec{b}$ are perpendicular. What is the angle between $\vec{a}$ and $\vec{b}$?

(a) π/6 (b) π/4
(c) π/3 (d) π/2

[NDA 2022 (II)]

143. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be unit vectors lying on the same plane. What is $\{(3\vec{a}+2\vec{b})\times(5\vec{a}-4\vec{c})\}.(\vec{b}+2\vec{c})$ equal to?

(a) –8 (b) –32
(c) 8 (d) 0

[NDA 2022 (II)]

144. What are the values of x for which the angle between the vectors
$2x^2\ \hat{i}+3x\hat{j}+\hat{k}$ and $\hat{i}-2\hat{j}+x^2\hat{k}$ is obtuse?

(a) 0<x<2 (b) x < 0
(c) x > 2 (d) 0≤ x ≤ 2

[NDA 2022 (II)]

145. The position vectors of vertices A, B and C of triangle ABC are respectively $\hat{j}+\hat{k}, 3\hat{i}+\hat{j}+5\hat{k}$ and $3\hat{j}+3\hat{k}$. What is angle C equal to?

(a) π/6 (b) π/4
(c) π/3 (d) π/2

[NDA 2022 (II)]

Consider the following for the next two (02) items that follow:

Let $\vec{a} = 3\hat{i}+3\hat{j}+3\hat{k}$ and $\vec{c} = \hat{j}-\hat{k}$. Let $\vec{b}$ be such that $\vec{a}.\vec{b} = 27$ and $\vec{a}\times\vec{b} = 9\vec{c}$.

146. What is $\vec{b}$ equal to?

(a) $3\hat{i}+4\hat{j}+2\hat{k}$ (b) $5\hat{i}+2\hat{j}+2\hat{k}$

(c) $5\hat{i}-2\hat{j}+6\hat{k}$ (d) $3\hat{i}+3\hat{j}+4\hat{k}$

[NDA – 2023 (1)]

147. What is the angle between $(\vec{a}+\vec{b})$ and $\vec{c}$?

(a) π/2 (b) π/3
(c) π/4 (d) π/6

[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:

Let a vector $\vec{a} = 4\hat{i}-8\hat{j}+\hat{k}$ make angles α, β, γ with the positive directions of x, y, z axes respectively.

148. What is cosα equal to?

(a) $\frac{1}{3}$ (b) $\frac{4}{9}$

(c) $\frac{5}{9}$ (d) $\frac{2}{3}$

[NDA – 2023 (1)]

149. What is cos2β + cos2 γ equal to?

(a) $-\frac{32}{81}$ (b) $-\frac{16}{81}$

(c) $\frac{16}{81}$ (d) $\frac{32}{81}$

[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:

The position vectors of two points A and B are $\hat{i}-\hat{j}$ and $\hat{j}+\hat{k}$ respectively.

150. Consider the following points:
1.(–1, –3, 1)
2.(–1, 3, 2)
3.(–2, 5, 3)
Which of the above points lie on the line joining A and B?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA – 2023 (1)]

151. What is the magnitude of $\overrightarrow{AB}$?

(a) 2 (b) 3
(c) √6 (d) $\sqrt{3}$

[NDA – 2023 (1)]

152. What is the length of projection of the vector $\hat{i}+2\hat{j}+3\hat{k}$ on the vector $2\hat{i}+3\hat{j}-2\hat{k}$?

(a) $\frac{1}{\sqrt{17}}$ (b) $\frac{2}{\sqrt{17}}$

(c) $\frac{3}{\sqrt{17}}$ (d) $\frac{2}{\sqrt{14}}$

[NDA-2023 (2)]

153. If $(\vec{a}\times\vec{b})^2+(\vec{a}.\vec{b})^2=144$ and $|\vec{b}|=4$, then what is the value of $|\vec{a}|$?

(a) 3 (b) 4
(c) 5 (d) 6

[NDA-2023 (2)]

154. If θ is the angle between vectors $\vec{a}$ and $\vec{b}$ such that $\vec{a}.\vec{b}\geq 0$, then which of the following is correct?

(a) $0\leq\theta\leq\pi$ (b) $\pi/2\leq\theta\leq\pi$
(c) $0\leq\theta\leq\pi/2$ (d) $0<\theta<\pi/2$

[NDA-2023 (2)]

155. The vectors $60\hat{i}+3\hat{j}$, $40\hat{i}-8\hat{j}$ and $\beta\hat{i}-52\hat{j}$ are collinear if:

(a) β=20 (b) β=40
(c) β=–40 (d) β=26

[NDA-2023 (2)]

156. Consider the following in respect of the vectors $\vec{a}=(0,1,1)$ and $\vec{b}=(1,0,1)$:

1.The number of unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$ is only one

2.The angle between the vectors is π/3

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2023 (2)]

157. Let $\vec{a}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$. If $\vec{a}\times(\vec{b}\times\vec{a})=\alpha\hat{i}-\beta\hat{j}+\gamma\hat{k}$, then what is the value α + β + γ

(a) 8 (b) 7
(c) 6 (d) 1

[NDA-2024 (1)]

158. If a vector of magnitude 2 units makes an angle $\frac{\pi}{3}$ with $2\hat{i}$, $\frac{\pi}{4}$ with $3\hat{j}$ and an acute angle θ with $4\hat{k}$, then what are the components of the vector?

(a) $(1,\sqrt{2},1)$ (b) $(1,-\sqrt{2},1)$
(c) $(1,-\sqrt{2},-1)$ (d) $(1,\sqrt{2},-1)$

[NDA-2024 (1)]

159. Consider the following in respect of moment of a force?:

1. The moment of force about a point is independent of point of application of force

2. The moment of a force about a line in a vector quantity

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA-2024 (1)]

160. For any vector $\vec{r}$, what is $(\vec{r}.\hat{i})(\vec{r}\times\hat{i})+(\vec{r}.\hat{j})(\vec{r}\times\hat{j})+(\vec{r}.\hat{k})(\vec{r}\times\hat{k})$ equal to ?

(a) $\vec{0}$ (b) $\vec{r}$
(c) $2\vec{r}$ (d) $3\vec{r}$

[NDA-2024 (1)]

161. Let $\vec{a}$ and $\vec{b}$ are two vectors of magnitude 4 inclined at angle $\frac{\pi}{3}$, then what is the angle between $\vec{a}$ and $\vec{a}-\vec{b}$?

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

[NDA-2024 (1)]

162. What is 3α + 2β equal to if $(2\hat{i}+6\hat{j}+27\hat{k})\times(\hat{i}+\alpha\hat{j}+\beta\hat{k})$ is a null vector ?

(a) 36 (b) 33
(c) 30 (d) 27

[NDA-2024 (2)]

163. For what value of angle between the vectors $\vec{a}$ and $\vec{b}$ is the quantity $(\vec{a}\times\vec{b})+\sqrt{3}(\vec{a}.\vec{b})$ maximum ?

(a) 0° (b) 30°
(c) 45° (d) 60°

[NDA-2024 (2)]

164. Let θ be the angle between two unit vectors $\vec{a}$ and $\vec{b}$. If $\vec{a}+2\vec{b}$ is perpendicular to $5\vec{a}-4\vec{b}$, then what is cosθ + cos2θ equal to ?

(a) 0 (b) 1/2
(c) 1 (d) $\frac{\sqrt{3}+1}{2}$

[NDA-2024 (2)]

165. Let ABCDEF be a regular hexagon. If $\overrightarrow{AD}=m\overrightarrow{BC}$ and $\overrightarrow{CF}=n\overrightarrow{AB}$, then what is mn equal to ?.

(a) – 4 (b) – 2
(c) 2 (d) 4

[NDA-2024 (2)]

166. The vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are of the same length. If take pairwise, they form equal angles. If $\vec{a}=\hat{i}+\hat{j}$, and $\vec{b}=\hat{j}+\hat{k}$, then what can $\vec{c}$ be equal to ?

I. $\hat{i}+\hat{k}$ II. $\frac{(-\hat{i}+4\hat{j}-\hat{k})}{3}$

Select the correct answer using the code given below ?

(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA-2024 (2)]

167. A line makes angles α, β and γ with the positive directions of the three coordinate axes. If $\vec{a}=(\sin^2\alpha)\hat{i}+(\sin^2\beta)\hat{j}+(\sin^2\gamma)\hat{k}$ and $b=\hat{i}+\hat{j}+\hat{k}$ then what is $\vec{a}.\vec{b}$ equal to?

(a) – 2 (b) – 1
(c) 1 (d) 2

[NDA-2025 (1)]

168. Consider the following statements in respect of a vector $\vec{d}=(\vec{a}\times\vec{b})\times\vec{c}$

I. $\vec{d}$ is coplanar with $\vec{a}$ and $\vec{b}$

II. $\vec{d}$ is perpendicular to $\vec{c}$

Which of the statements given above is/are correct?

(a) I only (b) II only
(c) Both I and II (d) neither I nor II

[NDA-2025 (1)]

169. The position vectors of three points A, B, and C are $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively such that $3\vec{a}-4\vec{b}+\vec{c}=\vec{0}$. What is AB : BC equal to?

(a) 3 : 1 (b) 1 : 3
(c) 3 : 4 (d) 1 : 4

[NDA-2025 (1)]

170. The position vectors of three points A, B and C are $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively where $\vec{c}=\left(\cos^2\theta\right)\vec{a}+\left(\sin^2\theta\right)\vec{b}$

What is $\left(\vec{a}\times\vec{b}\right)+\left(\vec{b}\times\vec{c}\right)+\left(\vec{c}\times\vec{a}\right)$ equal to?

(a) $\vec{0}$ (b) $2\vec{c}$

(c) $3\vec{c}$ (d) Unit vector

[NDA-2025 (1)]

171. Let $\vec{a}, \vec{b}$, $\left(\vec{a}\times\vec{b}\right)$ be unit vectors. What is $\left(\vec{a}.\vec{b}\right)$ equal to?

(a) 0 (b) 1/2

(c) 1 (d) 3

[NDA-2025 (1)]

172. Let $\vec{p}=\vec{a}-\vec{b}, \vec{q}=\vec{a}+\vec{b}$. If $|\vec{a}|=|\vec{b}|=2$ and $\vec{a}.\vec{b}=2$, then what is the value of $|\vec{p}\times\vec{q}|$?

(a) √3 (b) √6

(c) 2√3 (d) 4√3

[NDA-2025 (2)]

173. How many of the following can be a vector perpendicular to both the vectors $2\hat{i}-\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+3\hat{k}$?

I. $4\hat{i}+5\hat{j}-3\hat{k}$

II. $-8\hat{i}-10\hat{j}+6\hat{k}$

III. $\frac{1}{50}\left(-4\hat{i}-5\hat{j}+3\hat{k}\right)$

Select the correct answer

(a) none (b) one

(c) two (d) All three

[NDA-2025 (2)]

174. What is the area of the parallelogram whose sides are represented by the vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $2\hat{i}+\hat{j}+2\hat{k}$?

(a) $\frac{1}{2}\sqrt{26}$ (b) $\frac{1}{2}\sqrt{27}$

(c) $\sqrt{26}$ (d) $\sqrt{27}$

[NDA-2025 (2)]

175. The position vectors of the vertices A, B, C and D of a quadrilateral ABCD are given by $3\hat{i}+4\hat{j}-2\hat{k}$, $4\hat{i}-4\hat{j}-3\hat{k}$, $2\hat{i}-3\hat{j}+2\hat{k}$ and $6\hat{i}-2\hat{j}+\hat{k}$ respectively. What is the angle between the diagonals AC and BD of the quadrilateral?

(a) 90° (b) 75°

(c) 60° (d) 45°

[NDA-2025 (2)]

176. A force $\vec{F}=2i-\lambda\hat{j}+5\hat{k}$ is applied at the point A(1,2,5). If its moment about the point B(− 1, −2, 3) is $16\hat{i}-6\hat{j}+2\lambda\hat{k}$, then what is the value of λ?

(a) – 2 (b) 0

(c) 1 (d) 2

[NDA-2025 (2)]

ANSWER KEY

1.	c	**2.**	b	**3.**	a	**4.**	c	**5.**	c	**6.**	b	**7.**	c	**8.**	b	**9.**	a	**10.**	d
11.	d	**12.**	c	**13.**	d	**14.**	c	**15.**	c	**16.**	c	**17.**	b	**18.**	c	**19.**	b	**20.**	c
21.	c	**22**	a	**23.**	b	**24.**	c	**25.**	d	**26.**	a	**27.**	d	**28.**	c	**29.**	d	**30.**	a
31.	d	**32.**	c	**33.**	b	**34.**	b	**35.**	b	**36.**	a	**37.**	d	**38.**	d	**39.**	a	**40.**	a
41.	d	**42.**	c	**43.**	a	**44.**	b	**45.**	c	**46.**	b	**47.**	d	**48.**	b	**49.**	a	**50.**	a
51.	c	**52.**	b	**53.**	a	**54.**	a	**55.**	a	**56.**	c	**57.**	b	**58.**	d	**59.**	a	**60.**	d
61.	a	**62.**	d	**63.**	b	**64.**	c	**65.**	a	**66.**	c	**67.**	b	**68.**	b	**69.**	a	**70.**	c
71.	b	**72.**	d	**73.**	c	**74.**	b	**75.**	c	**76.**	B	**77.**	a	**78.**	c	**79.**	b	**80.**	a
81.	b	**82.**	c	**83.**	b	**84.**	c	**85.**	a	**86.**	d	**87.**	d	**88.**	a	**89.**	b	**90.**	b
91.	b	**92.**	c	**93.**	b	**94.**	a	**95.**	c	**96.**	b	**97.**	b	**98.**	b	**99.**	d	**100.**	a
101.	a	**102.**	a	**103.**	a	**104.**	c	**105.**	c	**106.**	c	**107.**	d	**108.**	a	**109.**	d	**110.**	a
111.	b	**112.**	d	**113.**	a	**114.**	b	**115.**	d	**116.**	b	**117.**	b	**118.**	c	**119.**	b	**120.**	b
121.	a	**122.**	c	**123.**	c	**124.**	a	**125.**	b	**126.**	c	**127.**	b	**128.**	c	**129.**	c	**130.**	d
131.	b	**132.**	d	**133.**	a	**134.**	a	**135.**	c	**136.**	c	**137.**	c	**138.**	c	**139.**	c	**140.**	d
141.	d	**142.**	c	**143.**	d	**144.**	a	**145.**	d	**146.**	b	**147.**	a	**148.**	b	**149.**	a	**150.**	b
151.	c	**152.**	b	**153.**	a	**154.**	c	**155.**	c	**156.**	b	**157.**	*	**158.**	a	**159.**	b	**160.**	a
161.	b	**162.**	a	**163.**	b	**164.**	d	**165.**	a	**166.**	c	**167.**	d	**168.**	c	**169.**	b	**170.**	a
171.	a	**172.**	d	**173.**	d	**174.**	c	**175.**	a	**176.**	a								

Solutions

Sol.1. (c)
Let $x\hat{i}+y\hat{j}+z\hat{k}$ is a unit vector
$\therefore x^2+y^2+z^2=1$
Given $x:y+z=\sqrt{3}\ :2:3$
$\Rightarrow x=\sqrt{3}\ k, y=2k\ and\ \ z=3k$
$\therefore \left(\sqrt{3}\ k\right)^2+\left(\sqrt{2}\ k\right)^2+\left(3\ k\right)^2=1$
$\Rightarrow 3k^2+4k^2+9k^2=1$
$\Rightarrow k^2=\frac{1}{16}\Rightarrow k=\frac{1}{4}$
Hence, $z=3k=3\times\frac{1}{4}=\frac{3}{4}$

Sol.2. (b)
Let $\vec{a}=i-2j+\hat{k}$ and $\vec{b}=4i-4\hat{j}+7\hat{k}$
$\vec{a}\,.\,\vec{b}=4(1)+4(2)+1(7)=19$
$\left|\vec{b}\right|=\sqrt{(4)^2+(4)^2+(7)^2}=\sqrt{81}=9$
$\therefore$ Projection of $\vec{a}$ on $\vec{b}$
$=\frac{\vec{a}.\vec{b}}{\left|\vec{b}\right|}=\frac{19}{9}$

Sol.3. (a)
Let the vector $\vec{a}$ lies in the plane of vectors $\vec{b}$ and $\vec{c}$.
$\Rightarrow \vec{a}, \vec{b}$ and $\vec{c}$ are coplanar.
$\therefore\ \ \vec{a}.\left(\vec{b}\times\vec{c}\ \right)=0$

Sol.4. (c)
Let
$p=\text{Magnitude } of\ 3\hat{i}-2\hat{j}=\sqrt{9+4}=\sqrt{13}$
$q=\text{Magnitude } of\ 2\hat{i}+2\hat{j}+\hat{k}=\sqrt{4+4+1}=3$
$r=\text{Magnitude } of\ 4\hat{i}-\hat{j}+\hat{k}=\sqrt{16+1+1}=\sqrt{18}=3\sqrt{2}$
$s=\text{Magnitude } of\ 2\hat{i}+2\hat{j}+3\hat{k}=\sqrt{4+4+9}=\sqrt{17}$
$\therefore r>s>p>q$

Sol.5. (c)
unit vector perpendicular to the vectors
$4\hat{i}+2\hat{j}$ and $-3\hat{i}+2\hat{j}$ is cross product of both vectors, these two vectors are lie in XY plane so cross product will be in Z direction. unit vector in Z direction is $\hat{k}$

Sol.6. (b)
Since, $\vec{u}_4=2\vec{u}_4$
$\therefore \vec{u}_2$ is parallel to $\vec{u}_4$.

Sol.7. (c)
Since, the points with position vectors
$10\hat{i}+3\hat{j},\ 12\hat{i}-5\hat{j},\ m\hat{i}+11\hat{j}$ are collinear

$$\therefore\begin{vmatrix}10 & 3 & 1\\ 12 & -5 & 1\\ m & 11 & 1\end{vmatrix}=0$$

$\Rightarrow 10(-5-11)-3(12-m)+1(132+5m)=0$
$\Rightarrow 160-36+3m+132++5m=0$
$\Rightarrow -196+132+8m=0$
$\Rightarrow 8m=64$
$\Rightarrow m=8$

Sol.8. (b)
We know that, the angle between the vectors
$a_1\hat{i}+b_1\hat{j}+c_1\hat{k}$ and $a_2\hat{i}+b_2\hat{j}+c_2\hat{k}$ is given by

$$\cos\theta=\left[\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right]$$

$\therefore$ Angle between the vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+2\hat{j}+3\hat{k}$ is given by

$$\cos\theta=\left[\frac{1\times(-1)+2\times2+3\times3}{\sqrt{1+4+9}\sqrt{1+4+9}}\right]$$

$=\frac{-1+4+9}{14}=\frac{12}{14}=\frac{6}{7}$
Now, $\sin\theta=\sqrt{1-\cos^2\theta}$
$=\sqrt{1\frac{36}{49}}=\sqrt{\frac{49-36}{49}}=\sqrt{\frac{13}{49}}=\frac{\sqrt{13}}{7}$

Sol.9. (a)
Let $\vec{OA}=i+2j+3k,\ \vec{OB}=2i+5j-\hat{k}$ and $\vec{OC}=-i+\hat{j}+2\hat{k}$,
Now, $\vec{AB}=\vec{OB}-\vec{OA}-i+3\hat{j}-4\hat{k}$
And $\vec{AC}=\vec{OC}-\vec{OA}=-2\hat{i}+\hat{j}-\hat{k}$
$\therefore$ Area of $\Delta ABC=\frac{1}{2}\left|\vec{AB}\times\vec{AC}\right|$

$$\text{Now, }\vec{AB}\times\vec{AC}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 3 & -4\\ -2 & -1 & -1\end{vmatrix}=0$$

$=\hat{i}(-3-4)-j(-3-8)+\hat{k}-(-1+6)$
$=-7i+9i+5i$
Now, $\left|\vec{AB}\times\vec{AC}\right|=\sqrt{155}$
$\therefore$ Required Area$=\frac{\sqrt{155}}{2}$

Sol.10. (d)
Let $\vec{c}$ is the unit vector perpendicular to both the vectors $\vec{a}$ and $\vec{b}$.

So, A unit vector which is perpendicular to both the Vectors $\vec{a}$ and $\vec{b}$ is $\frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$

Sol.11. (d)
Given vectors $-\hat{i}-2x\hat{j}-3y\hat{k}$ and $\hat{i}-3x\hat{j}-2y\hat{k}$
$\therefore\left(-\hat{i}-2x\hat{j}-3y\hat{k}\right).\left(\hat{i}-3x\hat{j}-2y\hat{k}\right)=0$
$\Rightarrow(-1)(1)+(-2x)(-3x)+(-3y)(-2y)=0$
$\Rightarrow -1+6x^2+6y^2=0$
$\Rightarrow 6x^2+6y^2=1$
Hence, locus of (x,y) is a circle

Sol.12. (c)
Since, the points with position vectors $10\hat{i}+3\hat{j},\ 12\hat{i}-5\hat{j},\ m\hat{i}+11\hat{j}$ are collinear
$\therefore\begin{vmatrix}10 & 3 & 1\\ 12 & -5 & 1\\ m & 11 & 1\end{vmatrix}=0$
$\Rightarrow 10(-5-11)-3(12-m)+1(132+5m)=0$
$\Rightarrow 160-36+3m+132++5m=0$
$\Rightarrow -196+132+8m=0$
$\Rightarrow 8m=64$
$\Rightarrow m=8$

Sol.13. (d)
Since the three vectors are coplanar, so one of them is expressible as a linear combination of the other two
$\therefore(m,-1,2)=x(2,-3,4)+y(1,2,-1)$
$\Rightarrow 2x+y=m$ ...(i)
$-3x+2y=-1$ …(ii)
and $4x-y=2$ ..(iii)
on solving equation (ii) and (iii) we get
$x=\frac{3}{5}$ and $y\frac{2}{5}$
$\therefore$ from (i), $2\left(\frac{3}{5}\right)+\frac{2}{5}=m$
$\Rightarrow\frac{6}{5}+\frac{2}{5}=m\Rightarrow\frac{8}{5}=m$

Sol.14. (c)
$\vec{a}.\vec{b}=0\Rightarrow\vec{a}\perp\vec{b}$ …(i)
$\vec{a}\times\vec{b}=0\Rightarrow\vec{a}\parallel\vec{b}$ …(ii)
From (i) and (ii) it is clear that $\vec{a}=0$ or $\vec{b}=0$

Sol.15. (c)
Since both vectors are orthogonal $\therefore$ their dot product is zero.
$\therefore 1(1)+(-x)(x)+(-y)(y)=0$
$\Rightarrow 1-x^2-y^2=0$
$\Rightarrow x^2+y^2=0$
Which is a circle

Sol.16. (c)
Let $\vec{a}=(2,1,-1),\vec{b}=(1,-1,0),\ \vec{c}=(5,-1,1)$
$\therefore\vec{a}+\vec{b}+\vec{c}=(-2,1,-2)$
Let $\vec{n}=xi+yj+z\hat{k}$ be the unit vector which is to $(-2,1,-2)$ in the opposite direction.
$\therefore x^2+y^2+z^2=1$ and

$\therefore x^2+y^2+z^2=1$ and $\begin{vmatrix}2 & -1 & 1\\ 1 & 2 & -3\\ 3 & m & 5\end{vmatrix}=0$
$\Rightarrow x=-2y,y=y,z=-2y$
$x^2+y^2+z^2=1\Rightarrow 4y^2+y^2+4y^2=1\Rightarrow y=\pm\frac{1}{3}$
Hence, the Required vector
$\hat{n}=\frac{2}{3}\hat{i}-\frac{\hat{j}}{3}+\frac{2}{3}\hat{k}$

Sol.17. (b)
by taking 2nd option
$(\vec{a}+\vec{b}).(\vec{a}-\vec{b})=0$
$a^2-b^2=0$
$a^2=b^2$
a = b

Sol.18. (c)
Three vector $x_1\vec{a}+y_1\vec{b}+z_1\vec{c}$, $x_2\vec{a}+y_2\vec{b}+z_2\vec{c}$ and $x_3\vec{a}+y_3\vec{b}+z_3\vec{c}$ will be coplanar if
$\begin{vmatrix}x_1 & y_1 & z_1\\ x_2 & y_2 & z_2\\ x_3 & x_3 & z_3\end{vmatrix}=0$
Here, $x_1=2,y_1=-1,z_1=1$
$x_2=1,y_2=2,z_2=-3$
$x_3=3,y_3=m\ \ z_3=5$
$\therefore\begin{vmatrix}2 & -1 & 1\\ 1 & 2 & -3\\ 3 & m & 5\end{vmatrix}=0$
$\Rightarrow 2(10+3m)+1(5+9)+1(m-6)=0$
$\Rightarrow 7m+28=0\Rightarrow m=-4$

Sol.19. (b)
We have
$\cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}$ and $\sin\theta=\frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$
Given : $|\vec{a}|=10,|\vec{b}|=2,\vec{a}.\vec{b}=12$
$\therefore\cos\theta=\frac{12}{20}$ and $\sin\theta=\frac{|\vec{a}\times\vec{b}|}{20}$
Now, by squaring and adding, we get
$\sin^2\theta+\cos^2\theta=1$
$\Rightarrow\frac{|\vec{a}\times\vec{b}|}{400}+\frac{144}{400}=1\Rightarrow|\vec{a}\times\vec{b}|=256$
$\Rightarrow|\vec{a}\times\vec{b}|=16$

Sol.20. (c)
From the given vector we can conclude that
$A\left(-1,\frac{1}{2},4\right),B\left(1,\frac{1}{2},4\right),C\left(1,\frac{-1}{2},4\right),D\left(-1,\frac{1}{2},4\right)$
Length = AB=2, BC=1
Area= $AB\times BC=2$

Sol.21. (c)
Consider
$\overrightarrow{PO}+\overrightarrow{OQ}=\overrightarrow{QO}+\overrightarrow{OR}$
$\Rightarrow\overrightarrow{OQ}-\overrightarrow{QO}=-\overrightarrow{PO}+\overrightarrow{OR}$
$\Rightarrow\overrightarrow{OQ}+\overrightarrow{OQ}=\overrightarrow{OR}+\overrightarrow{OP}$
$\Rightarrow\overrightarrow{OQ}=\frac{-}{2}\overrightarrow{OP}+\overrightarrow{OR}$

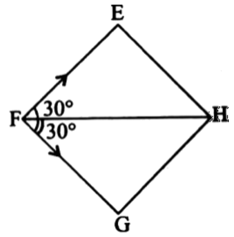
Hence Q is the mid-point of P and R.
$\therefore P,Q,R$ are collinear

Sol.22. (a)
Given
$\left(\lambda\hat{i}+\hat{j}-\hat{k}\right)\times\left(3\hat{i}-2\hat{j}+4\hat{k}\right)=\left(2\hat{i}-11\hat{j}-7\hat{k}\right)$
$\Rightarrow 2\hat{i}-\hat{j}(4\lambda+3)+\hat{k}(-2\lambda-3)=11\hat{j}-7\hat{k}$
$\Rightarrow -4\lambda-3=-11$
$\Rightarrow 4\lambda=8\Rightarrow\lambda=2$

Sol.23. (b)
The vector $2\hat{j}-\hat{k}$ lies in the plane of YZ

Sol.24. (c)
given vector lie in the plane of a and b vectors

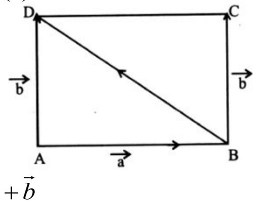
Sol.25. (d)



Rhombus $EFGH,\ \angle EFG=60^0$
$\angle EFG=30^0\ \angle EFG=30^0=\angle HFG$
From parallelogram of forces
$\overrightarrow{FE}+\overrightarrow{FG}=\overrightarrow{FG}$
Given $|\overrightarrow{FE}|+|\overrightarrow{FG}|=a(say)$
$\therefore\overrightarrow{FH}=2.\frac{\sqrt{3}}{2}\vec{a}=\sqrt{3}\ \vec{a}$
$\overrightarrow{EG}=\overrightarrow{EF}+\overrightarrow{EH}$
= a
$\sin 30^0+a\ \sin 30^0=a.\frac{1}{2}+a.\frac{1}{2}=a$
Thus, $\frac{\overrightarrow{FH}}{\overrightarrow{EG}}=\frac{\sqrt{3}a}{a}=\sqrt{3}$
So, $\overrightarrow{FH}=\sqrt{3}\ \overrightarrow{EG}$
$\Rightarrow m=\sqrt{3}$

Sol.26. (a)
Statement (1) : $4\hat{i}\times 3\hat{i}=0$
It is a true statement
$(\because\ \ )$
Statement (2) : $\frac{4\hat{i}}{3\hat{i}}=\frac{4}{3}$
It is an incorrect statement
$\because\ \frac{4\hat{i}}{3\hat{i}}=\frac{4}{3}\hat{i}$

Sol.27. (d)



$\overrightarrow{BD}=\vec{a}+\vec{b}$

Sol.28. (c)
$\vec{a}.\vec{b}=0\Rightarrow\vec{a}\perp\vec{b}$ …(i)
$\vec{a}\times\vec{b}=0\Rightarrow\vec{a}\parallel\vec{b}$ …(ii)
From (i) and (ii) it is clear that $\vec{a}=0$ or $\vec{b}=0$

Sol.29. (d)
Let $\vec{a}=p\left(-3i-2\hat{j}+13\hat{k}\right)$

$=(-3p)\hat{i}+(-2p)\hat{j}+(13p)\hat{k}$

It is given that $\vec{a}$ is of unit length

$\therefore |\vec{a}|=1 \Rightarrow |\vec{a}|=1$

$\Rightarrow(-3p))i(-3p)+(-2p)(-2p)+(13p)(13p)=1$

$9p^2+4p^2+169p^2=1$

$\Rightarrow p^2=\frac{1}{182} \Rightarrow p=\frac{1}{\sqrt{182}}$

Sol.30. (a)
Let $\vec{\alpha}=a\hat{i}+b\hat{j}+c\hat{k}$
Now,

$\vec{\alpha}.i=(\vec{a}i+\vec{b}j+\vec{c}k).i=\vec{a}$

$\vec{\alpha}.j=(\vec{a}i+\vec{b}j+\vec{c}k).j=\vec{b}$

$\vec{\alpha}.k=(\vec{a}i+\vec{b}j+\vec{c}k).k=\vec{c}$

Now, $\vec{a}.i+\vec{b}j+\vec{c}k=\vec{\alpha}$

Thus, Required expression = $\vec{\alpha}$

Sol.31. (d)
for normal vectors dot product should be zero.

Sol.32. (c)
Angle b/w the vectors is $\cos\theta=\frac{4(1)-4(1)}{\sqrt{32}\sqrt{3}}=0$

$\Rightarrow \theta=\frac{\pi}{2}$

Hence, $\cos\theta+\sin\theta=\cos\frac{\pi}{2}+\sin\frac{\pi}{2}=0+1=1$

Sol.33. (c)
Since magnitude of $\vec{a}\times\vec{b}$ = magnitude of $\vec{a}.\vec{b}$

$\therefore |\vec{a}|\times|\vec{b}|=|\vec{a}.\vec{b}|$

$\Rightarrow \therefore |\vec{a}|\times|\vec{b}|\sin\theta=|\vec{a}||\vec{b}|\cos\theta$

(By Definition)

$\Rightarrow \tan\theta=1$

$\Rightarrow \theta=\frac{\pi}{4}$

or any vector should be null vector

Sol.34. (b)
Consider $|\vec{a}+\vec{b}|^2+|\vec{a}-\vec{b}|^2=2(|\vec{a}|^2+|\vec{b}|^2)$

By putting the values of $|\vec{a}|,|\vec{b}|$ and $|\vec{a}+\vec{b}|$, we get

$6+|\vec{a}-\vec{b}|^2=2(2+3)$

$|\vec{a}-\vec{b}|=2$

Sol.35. (b)
Let $\vec{\beta}=a\hat{i}+b\hat{j}+c\hat{k}$

Since $\vec{\beta}.\vec{\alpha}=0$ and $\vec{\beta}.\vec{\gamma}=0$

$\therefore c=0 \quad 2a+3b=0 \Rightarrow a=-\frac{3b}{2}$

Hence, $\vec{\beta}=-\frac{3b}{2}\hat{i}+b\hat{j} \Rightarrow -3\hat{i}+2\hat{j}$

Sol.36. (a)
B(0, 0, 0) , A(1, 2, 3) and C(– 3, – 2, 1)
BA vector = $\hat{i}+2\hat{j}+3\hat{k}$
BC vector = $-3\hat{i}-2\hat{j}+\hat{k}$

$ABC=\frac{1}{2}|\overrightarrow{BA}\times\overrightarrow{BC}|$

= 3√5 sq units.

Sol.37. (d)

$\cos\frac{\pi}{3}=\frac{(\hat{i}-m\hat{j}).(\hat{j}+k)}{|\sqrt{1+m^2}||\sqrt{1^2+1^2}|}$

$\frac{1}{2}=\frac{-m}{\sqrt{1+m^2}.\sqrt{2}}$

$\frac{1}{2}=\frac{m^2}{1+m^2}$

$m=\pm1$

Sol.38. (d)

$(\hat{i}-2x\hat{j}-3y\hat{k}).(\hat{i}-3x\hat{j}-2y\hat{k})=0$

$1-6x^2-6y^2=1$

$x^2+y^2=\frac{1}{6}$

$x^2+y^2=\left(\sqrt{\frac{1}{6}}\right)^2$

Hence, locus of the point i.e. a circle

Sol.39. (a)

$|P(2\hat{i}-\hat{j}+2\hat{k})|=3$

$P\sqrt{|2^2+(-1)^2+2^2|}=3$

$3P=3 \Rightarrow 1$

Sol.40. (a)

$\vec{a}+t\vec{b}=(2-t)\hat{i}+(2+2t)\hat{j}+(3+t)\hat{k}$

$\vec{a}+t\vec{b}$ and $\vec{c}$ is perpendicular. Therefore,

$(\vec{a}+t\vec{b}).\vec{c}=0$

$3(2-t)+2+2t=0$

$6-3t+2t+2=0 \Rightarrow t=8$

Sol.41. (c)
Both vectors are in XY plane then unit vector along Z axis is perpendicular to both vectors so answer is $\hat{k}$

Sol.42. (c)

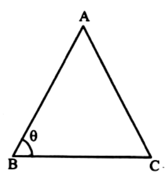
$\overrightarrow{AB}=2i+6\hat{j}-3\hat{k}$

$\overrightarrow{AB}=\sqrt{(-2)^2+6^2+(-3)^2}=\sqrt{49}=7$

Sol.43. (a)

$\overrightarrow{BA}=4\vec{i}+j+k$

$\overrightarrow{BC}=2\vec{i}+j+k$



$\cos B=\frac{\overrightarrow{BA}.\overrightarrow{BC}}{|\overrightarrow{BA}|.|\overrightarrow{BC}|}$

$=\frac{(4\hat{i}+\hat{j}+\hat{k}).(2\hat{i}-\hat{j}-\hat{k})}{|\sqrt{4^2+1^2+1^2}||\sqrt{2^2+(-1)^2+(-1)^2}|}=\frac{6}{\sqrt{18}\sqrt{6}}=\frac{1}{\sqrt{3}}$

Sol.44. (b)
Area of triangle $ABC=\frac{1}{2}|\overrightarrow{BA}\times\overrightarrow{BC}|$

$\overrightarrow{BA}\times\overrightarrow{BC}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 4 & 1 & 1\\ 2 & -1 & -1\end{vmatrix}$

$=\hat{i}(0)-\hat{j}(-6)+\hat{k}(-6)$

$|BA\times BC|=|6\hat{j}-6\hat{k}|=\sqrt{6^2+(-6)^2}$

$=6\sqrt{2}$

Area of triangle $=\frac{1}{2}\times6\sqrt{2}=3\sqrt{2}$

Sol.45. (c)
Mid-point of A and C,

$\left(\frac{2+0}{2},\frac{3+1}{2},\frac{1-1}{2}\right)=(1,2,0)$

Mid-point of B and C,

$\left(\frac{2+0}{2},\frac{3+1}{2},\frac{1-1}{2}\right)=\left(-1,\frac{3}{2},\frac{-1}{2}\right)$

Magnitude = $\sqrt{(1+1)^2+\left(2-\frac{3}{2}\right)^2+\left(\frac{1}{2}\right)^2}$

= $\sqrt{4+\frac{1}{4}+\frac{1}{4}}=\frac{3}{\sqrt{2}}$ units

Sol.46. (b)
Let angle between $\vec{a}$ and $\vec{b}$ be θ.

$|\vec{a}+\vec{b}|=\sqrt{|\vec{a}|^2+|\vec{b}|^2+2|\vec{a}||\vec{b}|\cos\theta|}$

$10\sqrt{3}=\sqrt{49+121+2\times7\times11\cos\theta}$

$300=170+154\cos\theta$

$154\cos\theta=130$

$|\vec{a}-\vec{b}|=\sqrt{|\vec{a}|^2+|\vec{b}|^2-2|\vec{a}||\vec{b}|\cos\theta}$

$|\vec{a}-\vec{b}|=\sqrt{170-154\cos\theta}$

$|\vec{a}-\vec{b}|=\sqrt{170-130}=\sqrt{40}$ or $2\sqrt{10}$

Sol.47. (d)
Let angle between $(\vec{a}+\vec{b})$ and $(\vec{a}-\vec{b})$ be α

$\cos\alpha=\frac{(\vec{a}+\vec{b})(\vec{a}-\vec{b})}{|\vec{a}+\vec{b}||\vec{a}-\vec{b}|}$

$=\frac{(7)^2-(11)^2}{10\sqrt{3}\times2\sqrt{10}}=\frac{(7+11)(7-11)}{20\sqrt{3}\times\sqrt{10}}=\frac{-18}{5\sqrt{30}}$

$=\frac{-6\times3}{5\sqrt{30}}\times\frac{\sqrt{30}}{\sqrt{30}}=\frac{3\sqrt{30}}{25}$

$\alpha=\cos^{-1}\left(\frac{-3}{5}\sqrt{\frac{6}{5}}\right)$

Sol.48. (b)
Projection of $\vec{a}$ on

$\vec{b}=\frac{\vec{a}.\vec{b}}{|\vec{b}|}=\frac{(\hat{i}-2\hat{j}+\hat{k}).(4\hat{i}-4\hat{j}+7\hat{k})}{|\sqrt{4^2+(-4)^2+7^2}|}$

$=\frac{19}{9}$

Sol.49. (a)
Vector perpendicular to $\vec{a}$ and $\vec{b}=\vec{a}\times\vec{b}$

$=\begin{bmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -2 & 1\\ 4 & -4 & 7\end{bmatrix}=\hat{i}(-14+4)-j(7-4)+\hat{k}(-4+8)$

$=10\hat{i}-3\hat{j}+4\hat{k}$ m

Sol.50. (a)
Given $|a|=2,|b|=5$ and $|a\times b|=8$

Also $|a\times b|=|a|.|b|.|\sin\theta|$

$\Rightarrow|\sin\theta|=\frac{8}{2\times5}=\frac{4}{5}$

$\Rightarrow |\cos\theta| = \frac{3}{5} \Rightarrow \cos\theta \pm \frac{3}{5}$

$\therefore a.b = |a|.|b|\cos\theta = 2\times5\times\frac{3}{5} = 6$

Sol.51. (c)

since $|a+b| = |a-b|$

$\Rightarrow [|a+b|]^2 = [a-b]^2$

$\Rightarrow a.a + b.b + a.b + b.a = a.a + b.b - a.b - b.a$

$\Rightarrow 4a.b = 0$

$\Rightarrow a.b = 0$

Hence, a is perpendicular to b.

Sol.52. (b)

Area of $\Delta OAB = \frac{1}{2}\left(\overline{OA}\times\overline{OB}\right)$

$\therefore \overline{OA}\times\overline{OB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ 2 & 1 & -3 \end{vmatrix}$

$= \hat{i}[3-1] - \hat{j}[-9-2] + \hat{k}[3+2]$

$= 2\hat{i} + 11\hat{j} + 5\hat{k}$

$\therefore |\overline{OA}\times\overline{OB}| = \sqrt{2^2+11^2+5^2} = \sqrt{150} = 5\sqrt{6}$

$\therefore$ Required area $= \frac{1}{2}\times5\sqrt{6} = \frac{5\sqrt{6}}{2}$ sq. units

Sol.53. (a)

According to question $\vec{a} = i + \vec{j} + k$

Then, $\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}$

$= \hat{i}[1+1] - [-1-1] + \hat{k}[1-1]$

$= 2\hat{i} + 2\hat{i} + 0 = 2\left(\hat{i}+\hat{j}\right)$

and $|\vec{a}\times\vec{b}| = \sqrt{4+4} = 2\sqrt{2}$

$\therefore$ required unit vector $= \pm\frac{2\left(\hat{i}+\hat{j}\right)}{2\sqrt{2}} = \pm\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Sol.54. (a)

Let $\vec{a} = \frac{-}{\sqrt{2}}i + \frac{1}{\sqrt{2}}\hat{j} + \hat{k}$

and $\vec{b} = \frac{-}{\sqrt{2}}i + \frac{1}{\sqrt{2}}\hat{j} + \hat{k}$

$\therefore \cos\theta = \frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}$

$= \frac{\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\hat{j} + \hat{k}\right)}{\sqrt{\frac{1}{2}+\frac{1}{2}+1}\sqrt{\frac{1}{2}+\frac{1}{2}+1}}$

$= \frac{1}{2}\left[\frac{1}{2} - \frac{1}{2} + 1\right] = \frac{1}{2} = \cos\, 60^0$

$\therefore \theta = 60^0$

Sol.55. (a)

Let $\vec{a} = \lambda i + (1+\lambda)\vec{j} + (1+2\lambda)k$

and $\vec{b} = \lambda i = (1+\lambda)\hat{i} + \lambda j + 2\hat{k}$

for a and b to be perpendicular, we should have

$\vec{a}.\vec{b} = |\vec{a}||\vec{b}|\cos\frac{\pi}{2} = 0$

$\Rightarrow \left[\lambda\hat{i} + (1+\lambda)\hat{j} + (1+2\lambda)\hat{k}\right].$

$\left[(1+\lambda)\hat{i} + \lambda\hat{j} + 2\hat{k}\right] = 0$

$\Rightarrow \lambda - \lambda^2 + \lambda + \lambda^2 + 2 + 4\lambda = 0$

$\Rightarrow 6\lambda = -2$

$\Rightarrow \lambda = \frac{2}{6} = -\frac{1}{3}$

Sol.56. (c)

Since $a + b + c = 0$

$\Rightarrow \vec{a} + \vec{b} = -\vec{c}$

$\Rightarrow |\vec{a}+\vec{b}| = |-\vec{c}| = |\vec{c}|$

$\Rightarrow |\vec{a}+\vec{b}| = |\vec{c}|$

$\Rightarrow |\vec{a}| + |\vec{b}| + 2\left(\vec{a}.\vec{b}\right) = |\vec{c}|$

$\Rightarrow (3)^2 + (5)^2 + 2\left(\vec{a}.\vec{b}\right) = (7)^2$

$\Rightarrow 2\left(\vec{a}.\vec{b}\right) = 15$

$\Rightarrow \vec{a}.\vec{b} = 15/2$

$\Rightarrow \cos\theta = \frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} = \frac{7.5}{3\times5} = \frac{1}{2}$

$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) \Rightarrow \theta = 60^0$

Sol.57. (b)

We have $a + b + c = 0$

So $|\vec{a}+\vec{b}+\vec{c}| = \vec{0}$

$|\vec{a}+\vec{b}+\vec{c}| = |\vec{a}| + |\vec{b}| + |\vec{c}| + 2\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right)$

$\Rightarrow 0 = (3)^2 + (5)^2 + (7)^2 + 2\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right)$

$\Rightarrow 0 = 9 + 25 + 49 + 2\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right)$

$\Rightarrow -\frac{83}{2} = \vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$

Sol.58. (d)

Since $a + b + c = 0$

$\Rightarrow \vec{b} + \vec{c} = -\vec{a}$

$\Rightarrow |\vec{b}+\vec{c}| = |-\vec{a}| = |\vec{a}|$

$\Rightarrow |\vec{b}+\vec{c}| = |\vec{a}|$

$\Rightarrow |\vec{b}| + |\vec{c}| + 2\left(\vec{b}.\vec{c}\right) = |\vec{a}|$

$\Rightarrow (5)^2 + (7)^2 + 2\left(\vec{b}.\vec{c}\right) = (3)^2$

$\Rightarrow 2\left(\vec{b}.\vec{c}\right) = -65$

$\Rightarrow \vec{b}.\vec{c} = -65/2$

$\Rightarrow \cos\theta = \frac{\vec{b}.\vec{c}}{|\vec{b}||\vec{c}|} = \frac{-65/2}{5\times7} = \frac{-13}{14}$

Sol.59. (a)

Since $a + b + c = 0$

$\Rightarrow \vec{a} + \vec{b} = -\vec{c}$

$\Rightarrow |\vec{a}+\vec{b}| = |-\vec{c}| = |\vec{c}| = 7$

Sol.60. (d)

Area of $\Delta ABC = \frac{1}{2}\left(\overline{AB}\times\overline{AC}\right)$

$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 3 & 2 \\ -4 & 5 & 2 \end{vmatrix}$

$= \frac{1}{2}\left[\hat{i}(6-10) - \hat{j}(-4+8) + \hat{k}(-10+12)\right] = 3$

square units

$\therefore$ Option (d) is correct.

Sol.61. (a)

Moment of force, $m = \vec{r}\times\vec{F}$

$m = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & -3 \\ 3 & 4 & -3 \end{vmatrix}$

$= \hat{i}(6+12) - \hat{j}(-6+9) + \hat{k}(8+6)$

$= 18\hat{i} - 3\hat{j} + 14\hat{k}$

$\sqrt{(18)^2 + (-3)^2 + (14)^2}$

$= \sqrt{529} = 23$ units.

Sol.62. (d)

$|\vec{a}| = 7, |\vec{b}| = 11$ and $|\vec{a}+\vec{b}| = 10\sqrt{3}$

Now $|\vec{a}+\vec{b}| = |\vec{a}| + |\vec{b}| + 2|\vec{a}||\vec{b}|\cos\theta$

$\therefore \left(10\sqrt{2}\right)^2 = 49 + 121 + 2\times7\times7\cos\theta$

$\therefore 300 = 170 + 154\cos\theta$

$\frac{300-170}{154} = \cos\theta$

$\therefore \frac{65}{77} = \cos\theta$

Now, $|\vec{a}-\vec{b}| = |\vec{a}| + |\vec{b}| - 2|\vec{a}||\vec{b}|\cos\theta$

$= (7)^2 + (11)^2 - 2\times7\times11\times\frac{65}{77}$

$= 49 + 121 - 2\times65$

$= 170 - 130 = 40$

$\therefore |\vec{a}-\vec{b}| = \sqrt{40} = 2\sqrt{10}$

$\therefore$ Option (d) is correct

Sol.63. (b)

α, β and γ be distinct real numbers

$\vec{a} = \alpha i + \beta\hat{j} + \gamma\hat{k}$

$\vec{b} = \beta i + \gamma\hat{j} + \alpha\hat{k}$

$\vec{c} = \gamma i + \alpha\hat{j} + \beta\hat{k}$

Let $\alpha = 1, \beta = 2$ and $\gamma = 3$

then, $\vec{b} = 2i + 3\hat{j}$

$\vec{c} = 3i + 2\hat{j} + 2\hat{k}$

Now,

$|\overline{ab}| = \sqrt{(2-1)^2 + +(3-2)^2 + +(1-3)^2} = \sqrt{6}$

Similarly $|\overline{bc}| = |\overline{ac}| = \sqrt{6}$

Hence, point for equilateral triangle.

Sol.64. (c)

If $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$

We know that when $\left(\vec{a}+\vec{b}\right).\left(\vec{a}-\vec{b}\right) = 0$, then $\vec{a}$ is perpendicular to $\vec{b}$

$= \vec{a}.\vec{a} - \vec{a}.\vec{b} + \vec{b}.\vec{a} - \vec{b}.\vec{b} = 0$

$\vec{a}$ is perpendicular to $\vec{b}$

$\therefore$ Option(c) is correct.

Sol.65. (a)

$\vec{u} - \vec{v} = \vec{w}$

$\left(2x\vec{\alpha} + y\beta\right) - \left(2y\vec{\alpha} + 3x\vec{\beta}\right) = 2\vec{\alpha} - 5\vec{\beta}$

$\therefore 2x-2y=2$ ...(i)

and $3x-y=5$ ...(ii)

solving equations (i) and (ii), we get

$x=2$ and $y=1$

Sol.66. (c)

Sol.67. (b)

If three vectors are co-planar.

$\Rightarrow\begin{vmatrix}\alpha & \alpha & \gamma\\ 1 & 0 & 1\\ \gamma & \gamma & \beta\end{vmatrix}=0$

$\Rightarrow \alpha[0-\gamma]-\alpha[\beta-\gamma]+[\gamma-0]=0$

$\Rightarrow -\alpha\gamma-\alpha\beta+\alpha\gamma+\gamma^2=0$

$\Rightarrow \gamma^2=\alpha\beta$

So α, γ, β are in G.P.

Sol.68. (b)

Length of diagonal

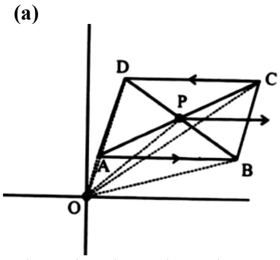
$=D=\sqrt{3^2+4^2}$

$\Rightarrow D=5$

$\therefore$ Area $=\frac{1}{2}(D)^2=\frac{25}{2}$

$=12.5$ *units*

Sol.69. (a)



$\because \overrightarrow{\quad}\ \overrightarrow{P}-\overrightarrow{AP};\overrightarrow{OB}=\overrightarrow{OP}-\overrightarrow{DP};$

$\overrightarrow{OC}=\overrightarrow{OP}+\overrightarrow{PC};\ \&\overrightarrow{OD}=\overrightarrow{OP}-\overrightarrow{DP};$

Also, $\overrightarrow{AP}=\overrightarrow{PC};\ \&\overrightarrow{DP}=\overrightarrow{PB}$

$\therefore \overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}=4\overrightarrow{OP}$

Sol.70. (c)

$\overrightarrow{OB}=\vec{b}$

$\overrightarrow{OC}=\vec{c}$

$\overrightarrow{OD}-\overrightarrow{OB}=4(\overrightarrow{OC}-\overrightarrow{OB})$

$\overrightarrow{OD}=4(\overrightarrow{OC})-3(\overrightarrow{OB})=4\vec{c}-3\vec{b}$

Sol.71. (b)

$\sqrt{(5-0)^2+(n-0)^2}=13$

$25+n^2=169$

$n^2=144$

$n=\pm 12$

Sol.72. (d)

$a^2b^2\sin^2\theta+a^2b^2\cos^2\theta=a^2b^2=4\times 9=36$

Sol.73. (c)

$\left|\hat{a}+\hat{b}\right|\le|\hat{a}|+\left|\hat{b}\right|$

$\left|\hat{a}\text{-}\hat{b}\right|\ge|\hat{a}|-\left|\hat{b}\right|$

Both are correct.

Sol.74. (b)

$\left|\hat{a}\text{-}\hat{b}\right|=\sqrt{3}\Rightarrow\left|\hat{a}\text{-}\hat{b}\right|=\left(\sqrt{3}\right)^2$

$\Rightarrow\left|\hat{a}\text{-}\hat{b}\right|^2=\left(\sqrt{3}\right)^2$

$\Rightarrow \hat{a}.\hat{a}+\hat{b}.\hat{b}-2\hat{a}.\hat{b}=3$

$\Rightarrow 2\hat{a}.\hat{b}=-1$

Now ; $\Rightarrow\left|\hat{a}+\hat{b}\right|=\hat{a}.\hat{a}+\hat{b}.\hat{b}+2\hat{a}.\hat{b}=1+1-1$

$\Rightarrow\left|\hat{a}+\hat{b}\right|=1$

$\Rightarrow\left|\hat{a}+\hat{b}\right|=1$

Sol.75. (c)

$\left(\vec{a}-\vec{d}\right)\times\left(\vec{b}-\vec{c}\right)$

$=\vec{a}\times\vec{b}-\vec{d}\times\vec{b}-\vec{a}\times\vec{c}+\vec{d}\times\vec{c}$

$=\vec{c}\times\vec{d}-\vec{d}\times\vec{b}-\vec{b}\times\vec{d}-\vec{c}\times\vec{d}$

$=-\vec{d}\times\vec{b}\times\vec{d}\times\vec{b}=0$

Again $=\left(\vec{a}\times\vec{b}\right)=\left(\vec{c}\times\vec{d}\right)$ given

$\Rightarrow\left(\vec{a}\times\vec{b}\right)\times\left(\vec{c}\times\vec{d}\right)=\left(\vec{c}\times\vec{d}\right)\times\left(\vec{c}\times\vec{d}\right)=0$

$\left(as\ \vec{a}\times\vec{a}=0\right)$

So both (1) and (2) are correct.

Sol.76. (b)

$\vec{A}=i+\hat{j}+\hat{k}$

$\vec{B}=2i+3\hat{j}-\hat{k}$

$\vec{A}\times\vec{B}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 1 & 1\\ 2 & 3 & -1\end{vmatrix}$

$=\hat{i}\ (-1-3)-\hat{j}(-1-2)+\hat{k}(3-2)$

$=4\hat{i}+3\hat{j}+\hat{k}$

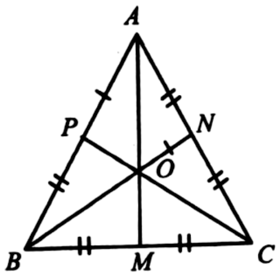
Vector of uni9t length orthogonal to both the vectors $\vec{A}$ and $\vec{B}$

$=\frac{\vec{A}\times\vec{R}}{\left|\vec{A}\times\vec{B}\right|}$

$=\frac{-4i+3j+k}{\sqrt{16+9+1}}=\frac{-4\hat{i}+3\hat{j}+\hat{k}}{\sqrt{26}}$

Sol.77. (a)

Position vectors of vertices A, B and C are $\vec{a}$, $\vec{b}$ and $\vec{c}$, .



$\because$ triangle is equilateral.

$\therefore$ Centroid and orthocenter will coincide

Centroid $\equiv$orthocenter position vector

$=\frac{1}{3}\left(\vec{a}+\vec{b}+\vec{c}\right)$

$\because$ given in question orthocenter is at origin.

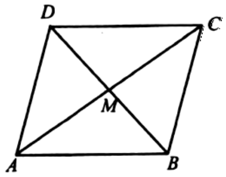
Hence $\frac{1}{3}\left(\vec{a}+\vec{b}+\vec{c}\right)=0$

$\boxed{\vec{a}+\vec{b}+\vec{c}=0}$

Sol.78. (c)

Diagonal d_1, $\overrightarrow{AC}=3i+j-2k$

Diagonal d_2, $\overrightarrow{BD}=i-3j+4k$



Area of parallelogram is $\frac{1}{2}\left|d_1\times d_2\right|$

Hence area = $\frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 1 & -2\\ 1 & -3 & 4\end{vmatrix}$

$=\frac{1}{2}\left|\left[\hat{i}(4-6)-\hat{j}(12+2)+\hat{k}(-9-1)\right]\right|$

$=\frac{1}{2}\left|-2\hat{i}-14\hat{j}-10\hat{k}\right|$

$=\frac{1}{2}\sqrt{4+196+100}$

$=\frac{10\sqrt{3}}{2}=5\sqrt{3}$ square units

Sol.79. (b)

$\vec{a}$ and $\vec{b}$ are two unit vectors.

$\because$ Hence,

$\left|\vec{a}+\vec{b}\right|=\left(\hat{a}+\hat{b}\right).\left(\hat{a}+\hat{b}\right)$

$=\hat{a}.\hat{a}+\hat{a}.\hat{b}+\hat{b}.\hat{a}+\hat{b}.\hat{b}$

$=\hat{a}.\hat{a}+\hat{b}.\hat{b}+2\hat{a}.\hat{b}$

$=\left|\hat{a}\right|^2+2\left|\hat{a}\right|\left|\hat{b}\right|\cos\theta+\left|\hat{b}\right|\left|\hat{b}\right|$

$=1+2\cos\theta+1\left[\because\ \hat{}\quad \text{unit vector}\right]$

$=2(1+\cos\theta)$

$\left|\hat{a}+\hat{b}\right|^2=2.2\cos^2\frac{\theta}{2}$

$\cos\frac{\theta}{2}=\frac{\left|\hat{a}+\hat{b}\right|}{2}$

Sol.80. (a)

$\left|\hat{a}-\hat{b}\right|^2=\left(\hat{a}-\hat{b}\right).\left(\hat{a}-\hat{b}\right)$

$=\hat{a}.\hat{a}+\hat{a}.\hat{b}+\hat{b}.\hat{a}+\hat{b}.\hat{b}$

$\left|\hat{a}\right|^2-2\left|\hat{a}\right|\left|\hat{b}\right|\cos\theta+\left|\hat{b}\right|^2$

$=2-2\cos\theta$

$=2\ (1-2\cos\theta)$

$\left|\hat{a}-\hat{b}\right|^2=2.2\sin^2\frac{\theta}{2}$

$\sin\frac{\theta}{2}=\frac{\left|\hat{a}-\hat{b}\right|}{2}$

Sol.81. (b)

We have $a+b+c=0$

So $\left|\vec{a}+\vec{b}+\vec{c}\right|=\vec{0}$

$\left|\vec{a}+\vec{b}+\vec{c}\right|=\left|\vec{a}\right|+\left|\vec{b}\right|+\left|\vec{c}\right|+2\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right)$

$\Rightarrow 0=(10)^2+(6)^2+(14)^2+2\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right)$

$\Rightarrow 0=100+36+196+2\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right)$

$\Rightarrow -166=\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$

Sol.82. (c)

Since $a+b+c=0$

$\Rightarrow \vec{a}+\vec{b}=-\vec{c}$

$\Rightarrow |\vec{a}+\vec{b}|=|-\vec{c}|=|\vec{c}|$

$\Rightarrow |\vec{a}+\vec{b}|\ =|\vec{c}|$

$\Rightarrow |\vec{a}|\ +|\vec{b}|\ +2|\vec{a}.\vec{b}|\ =|\vec{c}|$

$\Rightarrow (10)^2+(6)^2+2(\vec{a}.\vec{b})=(14)^2$

$\Rightarrow 2(\vec{a}.\vec{b})=60$

$\Rightarrow \vec{a}.\vec{b}=30$

$\Rightarrow \cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}=\frac{30}{10\times 6}=\frac{1}{2}$

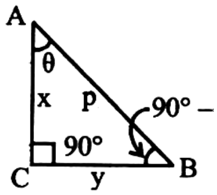
$\Rightarrow \theta=\cos^{-1}\left(\frac{1}{2}\right)\Rightarrow \theta=60^0$

Sol.83. (b)

$\overline{AB}.\overline{AC}+\overline{BC}.\overline{BA}+\overline{CA}.\overline{CB}$

$=(AB.AC.\cos\theta)+(BC.BA).\cos(90-\theta)$

$+(CA.CB.\cos 90)$



$=(p.x.\cos\theta)+(y.p.\sin\theta)+0$

$=[p.x.\cos\theta+y\sin\theta]$

By projection formula:

$p=x\cos\theta+y\cos(90-\theta)$

$=x\cos\theta+y\sin\theta$

$\therefore p[x\cos\theta+y\sin\theta]=p\times p=p^2$

Sol.84. (c)

Let point P is $(1,-1,2)$ and point Q is $(2,-1,3)$

$\Longrightarrow$ Position vector of P w.r.t. Q is

$\vec{r}=(1-2)\hat{i}+(-1+1)\hat{j}+(2-3)\hat{k}$

$\Longrightarrow$ Moment $=\vec{r}\times\vec{F}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -1 & 0 & -1\\ 3 & 2 & -4\end{vmatrix}$

$=\hat{i}(0+2)-\hat{j}(4+3)+\hat{k}(-2+0)=2\hat{i}-7\hat{j}-2\hat{k}$

Sol.85. (a)

Since $\vec{c}$ is parallel to $\vec{a}$

$\vec{c}=\lambda\vec{a}$

Now $\vec{b}=\vec{c}+\vec{d}=\lambda\vec{a}+\vec{d}$

$=\lambda(\hat{i}+\hat{j})+x\hat{i}+y\hat{j}+z\hat{k}$

$3\hat{i}+4\hat{k}=(\lambda+x)\hat{i}+(\lambda+y)\hat{j}+z\hat{k}$

Comparing we get

$z=4, \lambda+y=0,\ \lambda+x=3\Rightarrow -y+x=3$

.(from(1))

$\Rightarrow \lambda=-y$ ..(1) $\Rightarrow x-y=3$ ...(2)

Now $\vec{d}$ is $\perp r$ to $\vec{a}$

So, $\cos\theta=0$

$\Rightarrow x+y=0$ ...(3)

Solving (2) and (3) we get

$2x=3$

$\Rightarrow x=\frac{3}{2}, y=-\frac{3}{2}$

$\Rightarrow \vec{c}=\lambda(\vec{a})=\frac{-}{2}(i+\hat{j})$

Sol.86. (d)

Since $z=4$ and $x=\frac{3}{2}, y=-\frac{3}{2}$

So, neither 1 nor 2 is correct.

Sol.87. (d)

$\vec{a}=i-j+k,\ \vec{b}=2i+3j+2\hat{k}$

and $\vec{c}=i-mj+n\hat{k}; |\vec{c}|=\sqrt{6}$

Given, $\vec{a},\vec{b},\vec{c}$ are coplanar.

So, $\begin{vmatrix}1 & -1 & 1\\ 2 & 3 & 2\\ 1 & m & n\end{vmatrix}=0$

$c_1\to c_1+c_2 \qquad c_3\to c_3+c_2$

$\Rightarrow \begin{vmatrix}0 & 1 & 0\\ 5 & 3 & 5\\ 1+m & m & m-n\end{vmatrix}=0$

$\Rightarrow 0+1(5m+5n)-(5+5m)=0$

$5m+5n-5-5m=0\Rightarrow 5n=5\Rightarrow n=1$

$|\vec{c}|=6\Rightarrow \sqrt{1+m^2+n^2}=\sqrt{6}$

$\Rightarrow 1+m^2+n^2=6\Rightarrow 2+m^2=$

$6\Rightarrow m^2=4\Rightarrow m=\pm 2$

Sol.88. (a)

$\vec{a}\times\vec{b}=\vec{c},\quad \vec{b}\times\vec{c}=\vec{a}$

$\Rightarrow \vec{c}\perp\vec{a}$ & $\vec{c}\perp\vec{b}$

$\Rightarrow \vec{a}\perp\vec{b}\perp\vec{c}$

$\vec{a},\vec{b},\vec{c}$ are orthogonal

Now,

$|\vec{a}\times\vec{b}|=|\vec{c}|$

$\Rightarrow |\vec{a}||\vec{b}|\sin 90^o=|\vec{c}|$

$\Rightarrow |\vec{a}||\vec{b}|=|\vec{c}|\Rightarrow |\vec{b}||\cancel{\vec{c}}||\vec{b}|=|\cancel{\vec{c}}|$

$\Rightarrow |\vec{b}|\ =1$

$\Rightarrow |\vec{b}|=1$

$\vec{b}\times\vec{c}=\vec{a}$

$\Rightarrow \vec{a}\perp\vec{b}\perp\vec{c}$

Also, $|\vec{b}\times\vec{c}|=|\vec{a}|$

$|\vec{b}|\times|\vec{c}|\sin 90^o=|\vec{a}|$

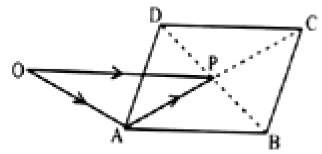
$\Rightarrow |\vec{b}|.|\vec{c}|=|\vec{a}|$

$\Rightarrow |\vec{c}|=|\vec{a}|$ (from (i))

Sol.89. (b)

From the figure, observe that

$\overline{OA}+\overline{AP}=\overline{OP}$



Similarly, $\overline{OB}+\overline{BP}=\overline{OP}$

$\overline{OC}+\overline{CP}=\overline{OP}$

$\overline{OD}+\overline{DP}=\overline{OP}$

$\Rightarrow \overline{OA}+\overline{OB}+\overline{OC}+\overline{OD}+\overline{AP}$

$+\overline{BP}+\overline{CP}+\overline{DP}=4\overline{OP}$

In a parallelogram diagonals bisect each other.

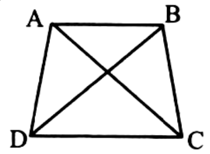
$\therefore \overline{AP}=\overline{PC} \qquad (\therefore\ \sin ce\ AP=CP)$

$\Rightarrow \overline{AP}=-\overline{PC}$

$\Rightarrow \overline{AP}+\overline{PC}=O$

$\therefore (i)\Rightarrow \overline{OA}+\overline{OB}+\overline{OC}+\overline{OD}=4\overline{OP}$

Sol.90. (b)



$\overline{BA}+\overline{AD}=\overline{BD}$ ....(i)

$\overline{CD}+\overline{DA}=\overline{CA}$ ...(ii)

$(i)+(ii)$

$\Rightarrow \overline{BA}+\overline{CD}=\overline{BD}+\overline{CA}\left(\because \overline{}\ \overline{}\ cancel\right)$

Sol.91. (b)

$\vec{a}=2i+3\hat{j}+4\hat{k}$ and

$\vec{b}=3i+2\hat{j}-\lambda\hat{k}$

Given, $\vec{a},\vec{b}$ are $\perp^r$

$\therefore \vec{a}.\vec{b}=0$

$\Rightarrow a_1a_2+b_1b_2+c_1c_2=0$

$\Rightarrow 6+6-4\lambda=0$

$\Rightarrow 4\lambda=12\Rightarrow \lambda=3$

Sol.92. (c)

We know,

$\cos^2\alpha+\cos^2\alpha+\cos^2\gamma=1$

$\Rightarrow \cos^2\alpha+\cos^2\beta=1-\cos^2 y=\sin^2\gamma\quad\therefore$

Statement 1 is correct.

Now, $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

$1-\sin^2\alpha+1-\sin^2\beta+1-\sin^2\gamma=1$

$\sin^2\alpha+\sin^2\beta+\sin^2\gamma=2$

$\therefore$ Statement 3 is correct.

Sol.93. (b)

$\vec{\alpha}=i+2\hat{j}-\hat{k}$

$\vec{\beta}=2i-\hat{j}+3\hat{k}$

$\vec{\gamma}=2i-j+6\hat{k}$

Let $\vec{\delta}=ai-b\hat{j}+c\hat{k}$

Since $\vec{\alpha}$ and $\vec{\delta}$ are perpendicular, $\vec{\alpha}.\vec{\delta}=0$

$\Rightarrow a+2b-c=0$ ...(1)

$\vec{\beta}$ and $\vec{\delta}$ are perpendicular, $\vec{\beta}.\vec{\delta}$ =0.

$\Rightarrow 2a-b+3c=0$ ...(2)

From

$(1),(2), \frac{a}{5}=\frac{b}{-5}=\frac{c}{-5} \Rightarrow \frac{a}{1}=\frac{b}{-1}=\frac{c}{-1}=x(say)$

So, a =x, b=x, c=-x

Also, it is given $\vec{\gamma}.\vec{\delta}=10$

$\Rightarrow 2a+b+6c=10$

$\Rightarrow 2x-x-6c=10$

$\Rightarrow -5x=10$

$\Rightarrow -x=-2$

$\therefore \vec{\delta}=2i+2\hat{j}+2\hat{k}$

$\therefore |\vec{\delta}|=\sqrt{4+4+4}=\sqrt{2}=2\sqrt{3}$

Sol.94. (a)

$(\hat{a}+\hat{b})\times(\hat{a}\times\hat{b})$

$=\hat{a}\times(\hat{a}\times\hat{b})+\hat{b}\times(\hat{a}\times\hat{b})$

$=(\hat{a}\cdot\hat{b})\hat{a}-(\hat{a}.\hat{a})\hat{b}+(\hat{b}\cdot\hat{b})\hat{b}$

$=\lambda\hat{a}-\hat{b}+\hat{a}\text{-}\lambda\hat{b}$

So, it is parallel to $\hat{a}-\hat{b}$

Sol.95. (c)

$\vec{F}=i+3\hat{j}+2\hat{k}$

$\overrightarrow{AB}=(3+1)\hat{i}+(-1-2)\hat{j}+(5-(-3))\hat{k}$

$2\hat{i}-3j+8\hat{k}$

Work done $=\vec{F}\cdot\overrightarrow{AB}$

$=(\hat{i}+3\hat{j}+2\hat{k}).(2\hat{i}-3\hat{j}+8\hat{k})$

$=(1\times2)+(3\times(-3))+(2\times8)$

$=2-9+16=9$ units.

Sol.96. (b)

Let $\vec{a}=xi+yj+z\hat{k}$

$|\vec{a}\times i|^2=y^2+z^2$

$|\vec{a}\times j|^2=x^2+z^2$

$|\vec{a}\times k|^2=x^2+z^2$

$\therefore |\vec{a}\times i|^2+|\vec{a}\times j|^2+|\vec{a}\times k|$

$=y^2+z^2+x^2+z^2+x^2+y^2$

$=2x^2+2y^2+2z^2$

$=2(x^2+y^2+z^2)$

$=2|\vec{a}|$

Sol.97. (b)

$a\hat{i}+\hat{j}+\hat{k}+b\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+c\hat{k}$

Are coplanar

$i.e., \begin{bmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{bmatrix}=0$

$c_2\to c_2-c_3; c_3\to c_3-c_1$

$\begin{vmatrix} a & 1\text{-}a & 1\text{-}a \\ 1 & b-1 & 0 \\ 1 & 0 & c-1 \end{vmatrix}=0$

$\Rightarrow a(b-1)(c-1)-(1-a)(c-1)-(1-a)(b-1)=0$

Divide both sides by $(1-a)(1-b)(1-c)$

$\Rightarrow \frac{a(b-1)(c-1)}{(1-a)(1-b)(1-c)}-\frac{(1-a)(c-1)}{(1-a)(1-b)(1-c)}$

$-\frac{(1-a)(b-1)}{(1-a)(1-b)(1-c)}=0$

$=\frac{a}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}=0$

$\Rightarrow \frac{1}{1-b}+\frac{1}{1-c}=-\frac{a}{1-a}$

Add $=\frac{1}{1-a}$ on both sides.

$\Rightarrow \frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}=\frac{1}{1-a}-\frac{a}{1-a}=1$

Sol.98. (b)

Given, $\vec{k}$ and $\vec{A}$ are parallel vectors. We know, cross product of parallel vectors is $\vec{o}$

Sol.99. (d)

$\vec{a}+2\vec{b}+3\vec{c}=\vec{0}$ ...(1)

$\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=\lambda(\vec{b}\times\vec{c})$ ...(2)

$(1)\times\vec{b}\Rightarrow \vec{b}\times\vec{a}+2\vec{b}\times\vec{b}+3\vec{b}\times\vec{c}=0$

$\Rightarrow \vec{a}\times\vec{b}+0+3\vec{b}\times\vec{c}=0$

$\Rightarrow \vec{a}\times\vec{b}=3\vec{b}\times\vec{c}$ ...(3)

$(1)\times\vec{c}\Rightarrow \vec{c}\times\vec{a}+2\vec{c}\times\vec{b}+3\vec{c}\times\vec{c}=0$

$\Rightarrow \vec{c}\times\vec{a}-2\vec{b}\times\vec{c}+0=0$ ...(4)

$\Rightarrow \vec{c}\times\vec{a}=2\vec{b}\times\vec{c}$

$\therefore (2)\Rightarrow \vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=\lambda(\vec{b}\times\vec{c})$

$\Rightarrow 3\vec{b}\times\vec{c}+\vec{b}\times\vec{c}+2(\vec{b}\times\vec{c})=\lambda(\vec{b}\times\vec{c})$

(from (3), (4)

$\Rightarrow 6(\vec{b}\times\vec{c})=\lambda(\vec{b}\times\vec{c})$

$\therefore \lambda=6$

Sol.100. (a)

$\vec{r}=(2i-j+3\hat{k})-(\hat{i}+2\hat{j}-\hat{k})=(\hat{i}+3\hat{j}-4\hat{k})$

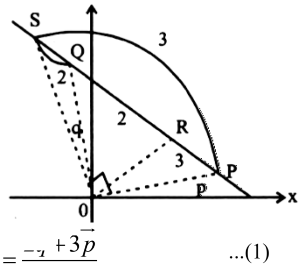
Moment $=(\tau)=\vec{r}\times\vec{f}$

$=(\hat{i}-3\hat{j}+4\hat{k})\times(3\hat{i}+\hat{k})$

$=-3\hat{i}+11\hat{j}+9\hat{k}$

Sol.101. (a)

R divides PQ internally in ratio 2:3



$\therefore \overrightarrow{OR}=\frac{2\vec{q}+3\vec{p}}{5}$ ...(1)

S divides PQ externally in ratio 2:3

$\overrightarrow{OS}=\frac{2\vec{q}-3\vec{p}}{2-3}=3\vec{p}-2\vec{q}$ ...(2)

Given, $\overrightarrow{OR}$ and $\overrightarrow{OS}$ are perpendicular.

$\therefore \frac{3\vec{p}+2\vec{q}}{5}(3\vec{p}-2\vec{q})=0$

$\Rightarrow 9p^2-4q^2=0\Rightarrow 9p^2=4q^2$

Sol.102. (a)

$|\vec{a}|=2, |\vec{b}|=7$

$\vec{a}\times\vec{b}=3i+2\hat{j}+6\hat{k}$

We know, $\vec{a}\times\vec{b}=|\vec{a}||\vec{b}|\sin\theta$

$\Rightarrow |3\hat{i}+2\hat{j}+6\hat{k}|=(2)(7)\sin\theta$

$\Rightarrow \sqrt{9+4+36}=(2)(7)\sin\theta$

$\Rightarrow \pm7=(2)(7)\sin\theta$

$\therefore \sin\theta\pm\frac{1}{2}$

$\sin\theta$ is acute angle, $\theta=30^0$

Sol.103. (a)

From triangle law of vector addition

$\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=0$

Only statement (1) is correct

Sol.104. (c)

$\vec{\tau}=\vec{r}\times\vec{F}=(i+2\hat{j}+3\hat{k})\times\lambda\hat{k}=2\lambda\hat{i}-\lambda\hat{j}$

$\Rightarrow |\vec{\tau}|=\sqrt{5}\lambda$

Sol.105. (c)

$(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})$

$=\vec{a}\times\vec{a}+\vec{a}\times\vec{b}-\vec{b}\times\vec{a}-\vec{b}\times\vec{b}$

$=0+\vec{a}\times\vec{b}+\vec{a}\times\vec{b}-0=2(\vec{a}\times\vec{b})$

Sol.106. (c)

For simplicity let us take $\vec{a},\vec{b},\vec{c}$ as $\hat{i},\hat{j},\hat{k}$ Now magnitude of $\vec{A}.\vec{B}$ and $\vec{C}$ will be $\sqrt{3}$

Sol.107. (d)

$|\vec{a}+\vec{b}|^2+|\vec{a}-\vec{b}|^2=2\{|\vec{a}|^2+|\vec{b}|^2\}=2\times25=50$

$\Rightarrow |\vec{a}+\vec{b}|^2+25=50\Rightarrow |\vec{a}+\vec{b}|^2+25$

$\Rightarrow |\vec{a}+\vec{b}|^2=5$

Sol.108. (a)

$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & -4 & -1 \end{vmatrix}$

$=\hat{i}(1+4)-\hat{j}(-2-3)+\hat{k}(-8+3)$

$=5\hat{i}+5\hat{j}-5\hat{k}$

Unit vector $=\frac{5}{5\sqrt{3}}\hat{i}+\frac{5}{5\sqrt{3}}\hat{j}-\frac{5}{5\sqrt{3}}\hat{k}$

$\frac{1}{\sqrt{3}}\hat{i}+\frac{1}{\sqrt{3}}\hat{j}-\frac{1}{\sqrt{3}}\hat{k}$

Sol.109. (d)

$(x\hat{i}+y\hat{j}+z\hat{k}).(\hat{i}+\hat{j}+\hat{k})=x+y+z$

Sol.110. (a)

$(\vec{a}+\vec{b}).(\vec{a}+\vec{b})=a^2+2\vec{a}.\vec{b}+b^2=a^2+b^2$

$\Rightarrow \vec{a}.\vec{b}=0\Rightarrow \vec{a}\perp\vec{b}$

Sol.111. (b)

$\vec{a}=i-2j-5\hat{k}$

$\vec{b}=2i+j-3\hat{k}$

$\vec{b}=\vec{a}=2i+j-3\hat{k}-(\hat{i}-2\hat{j}+5\hat{k})$

$= \hat{i}(2-1)+\hat{j}(1+2)+\hat{k}(-3-5)=\hat{i}+3\hat{j}-8\hat{k}$

$3\vec{a}+\vec{b}=3\left(i-2\hat{j}+5\hat{k}\right)+\left(2\hat{i}+\hat{j}-3\hat{k}\right)$

$=3\hat{i}-6\hat{j}+15\hat{k}+2\hat{i}+\hat{j}-3\hat{k}$

$=\hat{i}(3+2)+\hat{j}(-6+1)+k(15-3)$

$=5\hat{i}-5\hat{j}+12\hat{k}$

$\left(\vec{b}-\vec{a}\right).\left(3\vec{a}+\vec{b}\right)=\left(i+3\hat{j}-8\hat{k}\right).\left(5\hat{i}-5\hat{j}+12\hat{k}\right)$

$=(1)(5)+(3)(-5)+(-8)(12)$

$=5-15-96=-106$

Sol.112. (d)

Given, $\overline{OA}=3i-2\hat{j}+\hat{k}$

$\overline{OB}=2i-4\hat{j}+-3\hat{k}$

$\overline{AB}=\overline{OB}-\overline{OA}$

$=\left(2\hat{i}+4\hat{j}-3\hat{k}\right)-\left(3\hat{i}+2\hat{j}-\hat{k}\right)$

$=\hat{i}(2-3)+\hat{j}(4+2)+\hat{k}(-3-1)$

$=\hat{i}+6\hat{j}-4\hat{k}$

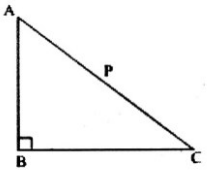
Length of AB= $\sqrt{(-1)^2+(6)^2+(-4)^2}$

$=\sqrt{1+36+16}=\sqrt{53}$

Sol.113. (a)

Hypotenuse, AC=P

BC is perpendicular to AB



$\therefore \overline{BC}.\overline{BA}=0$

$\therefore \overline{AB}.\overline{AC}+\overline{BC}.\overline{BA}+\overline{CA}.\overline{CB}$

$=\overline{AB}.\overline{AC}+\overline{AC}.\overline{BC}+\overline{AC}\left(\overline{AB}+\overline{BC}\right)$

$=\overline{AC}.\overline{AC}+\overline{AC}\quad=p^2$

Sol.114. (b)

Given, $\bar{a}=2\bar{i}-6\bar{j}-3\bar{k}$

$\bar{b}=4\bar{i}-3\bar{j}-\bar{k}$

$\bar{a}.\bar{b}=\left(2\bar{i}-6\bar{j}-3\bar{k}\right).\left(4\bar{i}+3\bar{j}-\bar{k}\right)$

$=(2)(4)+(-6)+(-3)(-1)$

$=8-18+3=-7$

$\Rightarrow \left|\vec{a}\right|\left|\vec{b}\right|\cos\theta=-7 \Rightarrow \cos\theta=\dfrac{-7}{\left|\vec{a}\right|\left|\vec{b}\right|}$

$=\dfrac{-7}{\sqrt{49}\sqrt{26}}=\dfrac{-1}{\sqrt{26}}$

$\sin\theta=\sqrt{1-\cos^2\theta}=\sqrt{1-\left(\dfrac{-1}{\sqrt{26}}\right)^2}=\sqrt{1-\dfrac{1}{26}}$

$=\sqrt{\dfrac{25}{26}}=\dfrac{5}{\sqrt{26}}$

Sol.115. (d)

Given vectors are $3\hat{i}+4\hat{j}-\hat{k}$ and $-2\hat{i}+\lambda\hat{j}+10\hat{k}$ If these are perpendicular, dot product is 0.

$\left(3\hat{i}+4\hat{j}-\hat{k}\right).\left(2\hat{i}+\lambda\hat{j}+10\hat{k}\right)=0$

$\Rightarrow (3)(-2)+(4)(\lambda)+(-10)=0$

$\Rightarrow 6+4\lambda-10=0$

$\Rightarrow 4\lambda=16$

$\Rightarrow \lambda=4$

Sol.116. (b)

Projection of $\vec{a}=i-2j+\hat{k}$, on $\vec{b}=4i-4j+7\hat{k}$

$=\dfrac{\vec{a}.\vec{b}}{\left|\vec{b}\right|}=\dfrac{\left(i-2\hat{j}+\hat{k}\right).\left(4\hat{i}-4\hat{j}+7\hat{k}\right)}{\sqrt{16+16+49}}=\dfrac{19}{9}$

Sol.117. (b)

If $\left|\vec{a}+\vec{b}\right|=\left|\vec{a}-\vec{b}\right|$

Taking square both sides

$\left|\vec{a}+\vec{b}\right|^2=\left|\vec{a}-\vec{b}\right|^2$

$\left(\vec{a}+\vec{b}\right).\left(\vec{a}+\vec{b}\right)=\left(\vec{a}-\vec{b}\right).\left(\vec{a}-\vec{b}\right)$

$=\vec{a}.\vec{a}+2\vec{a}.\vec{b}+\vec{b}.\vec{b}=\vec{a}.\vec{a}-2\vec{a}.\vec{b}+\vec{b}.\vec{b}$

$\Rightarrow \vec{a}.\vec{b}=0$

$\vec{a}$ is perpendicular to $\vec{b}$

Sol.118. (c)

1, 3 statement is correct.

Sol.119. (b)

Sol.120. (b)

$\left|\hat{a}-\hat{b}\right|^2=\left(\hat{a}-\hat{b}\right).\left(\hat{a}-\hat{b}\right)$

$=\hat{a}.\hat{a}+\hat{a}.\hat{b}+\hat{b}.\hat{a}+\hat{b}.\hat{b}$

$\left|\hat{a}\right|^2-2\left|\hat{a}\right|\left|\hat{b}\right|\cos\theta+\left|\hat{b}\right|^2$

$=2-2\cos\theta$

$=2\,(1-2\cos\theta)$

$\left|\hat{a}-\hat{b}\right|^2=2.2\sin^2\dfrac{\theta}{2}$

$\sin\dfrac{\theta}{2}=\dfrac{\left|\hat{a}-\hat{b}\right|}{2}$

$\sin^2\dfrac{\theta}{2}=\dfrac{\left|\hat{a}-\hat{b}\right|^2}{4}$

Sol.121. (a)

projection on x-axis=r cos α=cos30°= $\dfrac{\sqrt{3}}{2}$

projection on y-axis = rcosβ = cos60°= $\dfrac{1}{2}$

then vector will be = $r\cos\alpha\hat{i}+r\cos\beta\hat{j}$

$=\dfrac{\sqrt{3}}{2}\hat{i}+\dfrac{1}{2}\hat{j}$

Sol.122. (c)

projection on x-axis = 12cosα = cos45°= $12\dfrac{1}{\sqrt{2}}=6\sqrt{2}$

projection on y-axis = 12cosβ

= 12cos60°= 6

we know that $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

$\cos^2 45°+\cos^2 60°+\cos^2\gamma=1$

by solving it cosγ = $\pm\dfrac{1}{2}$

then vector will be

$=r\cos\alpha\hat{i}+r\cos\beta\hat{j}+r\cos\gamma\hat{k}$

$=6\sqrt{2}\hat{i}+6\hat{j}\pm6\hat{k}$

Sol.123. (c)

$\vec{a}=2i-4j+5\hat{k}$, $\vec{b}=i-2j-3\hat{k}$

diagonals

$\vec{a}+\vec{b}=3i-6j+2\hat{k}$

$\vec{a}-\vec{b}=i-2j+8\hat{k}$

$\left(\vec{a}+\vec{b}\right).\left(\vec{a}-\vec{b}\right)=\left(3i-6\hat{j}+2\hat{k}\right).\left(\hat{i}-2\hat{j}+8\hat{k}\right)=$

$3+12+16=31$

Sol.124. (a)

$a^2b^2\sin^2\theta+a^2b^2\cos^2\theta=a^2b^2=144$

$16\left|\vec{b}\right|^2=144$

$\left|\vec{b}\right|^2=9$

$\left|\vec{b}\right|=3$

Sol.125. (b)

if these vectors are coplanar then

$\begin{vmatrix}2 & -3 & 1\\ 1 & 2 & -3\\ 0 & 1 & p\end{vmatrix}=0$

by solving it p = −1

Sol.126. (c)

$\vec{r}=ai+bj$

if it is equally inclined to both x and y axes then α = β = 45°

$r\cos\alpha\hat{i}+r\cos\beta\hat{j}+r\cos\gamma\hat{k}$

$=2\cos45°\hat{i}+2\cos45°\hat{j}$

$=\sqrt{2}\hat{i}+\sqrt{2}\hat{j}$

a = √2 , b = √2

Sol.127. (b)

$\vec{c}=\vec{a}+\vec{b}$, where $\left|\vec{a}\right|=\left|\vec{b}\right|\neq0$

it means these three vectors are coplanar, and forming a triangle

$\vec{a}-\vec{b}$ will be also in same plane,

so statement I is wrong.

$\left(\vec{a}\times\vec{b}\right)$ is perpendicular to the plane

so statement II is correct.

Sol.128. (c)

$\left|\vec{a}+\vec{b}\right|=\left|\vec{a}-\vec{b}\right|$

$\Rightarrow \left|\vec{a}+\vec{b}\right|^2=\left|\vec{a}-\vec{b}\right|^2$

$\Rightarrow a^2+b^2+2\vec{a}.\vec{b}=a^2+b^2-2\vec{a}.\vec{b}$

$\Rightarrow \vec{a}.\vec{b}=0$

$\vec{a}$ must be perpendicular to $\vec{b}$

Sol.129. (c)

$\vec{a},\vec{b}$ and $\vec{c}$ are coplanar

then $\left(2\vec{a}\times3\vec{b}\right).4\vec{c}+\left(5\vec{b}\times3\vec{c}\right).6\vec{a}$

$24\left(\vec{a}\times\vec{b}\right).\vec{c}+90\left(\vec{b}\times\vec{c}\right).\vec{a}$

= 24 (0) + 90(0) = 0

Sol.130. (d)

1.The cross product of two unit vectors is always a unit vector

it is wrong because cross product also depends on angle.

2.The dot product of two unit vectors is always unit

it is wrong because dot product also depends on angle.

3.The magnitude of sum of two unit vectors is always greater than the magnitude of their difference.

It is wrong because $\left|\vec{a}+\vec{b}\right|$ can be less or more than $\left|\vec{a}-\vec{b}\right|$, it also depends upon angle

So, all statements are incorrect.

Sol.131. (b)

$\vec{a}\times\vec{b}$ is perpendicular to $\vec{C}$

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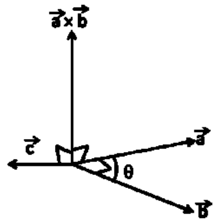


i.e. $\vec{a},\vec{b},\vec{c}$ lie in same plane

1. $\vec{a}\times\vec{b} \neq \sin\theta\ \vec{c}$

$\vec{a}\times\vec{b}$ is perpendicular of $\vec{c}$, so these vectors can not be parallel.

1. $\vec{a},\vec{b},\vec{c}$ are coplanar so $\vec{a}.(\vec{b}\times\vec{c})$=0

So only 2nd statement is correct?

Sol.132. (d)

$\vec{a}+3\vec{b}=\hat{\imath}-\hat{\jmath}$ …(1)

$2\vec{a}+\vec{b}=\hat{\imath}-2\hat{\jmath}$ …(2)

So solving above equation as linear equations

$\vec{a}=-\hat{\jmath} \quad \vec{b}=-\hat{\imath}$

$\vec{a}$ is towards the negative y axis

$\vec{b}$ is towards the positive x axis

Both vectors are perpendicular

Sol.133. (a)

$\vec{a}+\vec{b}$ is perpendicular of $\vec{a}$

$(\vec{a}+\vec{b}).\vec{a}=0$

$a^2+\vec{a}.\vec{b}=0$

$\vec{a}+\vec{b}=-a^2$

$(4\vec{a}+\vec{b}).\vec{b}=0$

$4\,\vec{a}.\vec{b}+b^2=0$

$4(-a^2)+\vec{b}.\vec{b}=0$ As $\vec{b}=2\,\vec{a}$

$4(-a^2)+(2\vec{a}).(2\vec{a})>0$

$-4a^2+4\,\vec{a}.\vec{a}$

$-4a^2+4\,a^2=0$

Sol.134. (a)

(1)$(\vec{a}.\vec{b})\times\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$

(2) $(\vec{a}.\vec{b})\times(\vec{c})$ is perpendicular to $\vec{a}$ and $\vec{b}$ first statement is true as we know the result if $\vec{a}$, $\vec{b}$ & $\vec{c}$ are coplanar then $(\vec{a}\times\vec{b})\times\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$.

(3) As $\vec{r}=(\vec{a}\times\vec{b})\times\vec{C}$

↓ ↓

Coplanar perpendicular

∴Statement is also correct.

Sol.135. (c)

$\overline{OA}=(\sqrt{2}-1)\,\hat{\imath}-\hat{\jmath}\rightarrow$P.V of A

$\overline{OA}=\hat{\imath}+(\sqrt{2}-1)\hat{\jmath}\rightarrow$ P.V. of B

$\overline{AB}=\overline{OB}-\overline{OA}$

$=\left[\hat{i}+(\sqrt{2}+1)\hat{j}\right]-\left\{\left[(\sqrt{2}-1)\hat{i}\right]-\hat{j}\right\}$

$=\left[1-\sqrt{2}+1\right]\hat{i}+\hat{j}\left[\sqrt{2}+1+1\right]$

$=(2\sqrt{2})\hat{i}+\hat{j}\left[2+\sqrt{2}\right]$

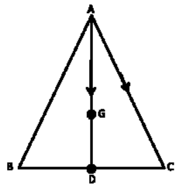
$(\overline{AB})=\sqrt{(2-\sqrt{2})^2+(2+\sqrt{2})^2}=2\sqrt{3}$

Sol.136. (c)

$4\hat{\imath}+\hat{\jmath}-3\hat{k}$ and $p\hat{\imath}+a\hat{\jmath}-2\hat{k}$ are collinear than

$\frac{4}{p}=\frac{1}{q}=\frac{-3}{-2}$ p= $\frac{8}{3}$ q= $\frac{2}{3}$

Sol.137. (c)



$\overrightarrow{AG}=\frac{2}{3}\overrightarrow{AD}$

$=\frac{2}{3}(\overrightarrow{AC}+\overrightarrow{CD})$

$=\frac{2}{3}\left(\overline{AC}+\frac{\overline{CB}}{2}\right)=\frac{2}{3}\left(\bar{c}-\bar{a}+\frac{\bar{b}-\bar{c}}{2}\right)$

$=\frac{2}{3}\left(\frac{\bar{c}-2\bar{a}+\bar{b}}{2}\right)=\frac{\bar{b}+\bar{c}-2\bar{a}}{3}$

Sol.138. (c)

1. $\bar{a}.(\bar{b}+\bar{c})=\bar{a}.\bar{b}.+\bar{a}.\bar{c}$ correct

2. $\bar{a}\times(\bar{b}+\bar{c})=\bar{a}.\times b.+\bar{a}\times\bar{c}$ correct

3. $\bar{a}\times(\bar{b}\times\bar{c})\neq\bar{a}.\times\bar{b}.\times\bar{c}$ third statement is incorrect

Sol.139. (c)

$\bar{a}.\times\bar{b}.=\bar{c}$

If any two vector are given then we can find a unique third vector by using above relation

Both statements are correct.

Sol.140. (d)

$|\bar{a}-\bar{b}|<2$

$\sqrt{a^2+b^2-2ab\cos 2\theta}<2$

$\sqrt{1+1-2\cos 2\theta}<2$

$\sqrt{2(1-\cos 2\theta)}<2$

$\sqrt{2\times 2\sin^2\theta}<2$

$|2\sin\theta|<2$

$|\sin\theta|<1$

$-1<\sin\theta<1$

Sol.141. (d)

Let O is the intersection point of diagonals

If $\overrightarrow{PR}=\vec{a}$, then $\overrightarrow{PO}=\frac{\vec{a}}{2}$

$\overrightarrow{QS}=\vec{b}$, then $\overrightarrow{QO}=\frac{\vec{b}}{2}$

now in triangle PQO

$\overrightarrow{PQ}+\overrightarrow{QO}=\overrightarrow{PO}$

$\overrightarrow{PQ}=\frac{\vec{a}-\vec{b}}{2}$

Sol.142. (c)

$\bar{a}+2\,\bar{b}$ and $5\bar{a}-4\vec{b}$ are perpendicular

than

$(\vec{a}+2\vec{b}).(5\vec{a}-4\vec{b})=0$

$(\vec{a}+2\vec{b}).(5\vec{a}-4\vec{b})=5a^2+10\vec{b}.\vec{a}-4\vec{a}\vec{b}-8b^2=0$

$5+6ab\cos\theta-8=0$

$\cos\theta=\frac{1}{2}$

$\theta=60°$

Sol.143. (d)

$\{(3\bar{a}+2\vec{b})\times(5\bar{a}-4\vec{c})\}.(\bar{b}+2\vec{c})=0$

Sol.144. (a)

If angle between two vectors is obtuse than their dot product will be negative.

$(2x^2\hat{i}+3x\hat{j}+\hat{k}).(\hat{i}-2\hat{j}+x^2\hat{k})<0$

$2x^2-6x+x^2<0$

$x^2-2x<0$

$x(x-2)<0$

$0<x<2$

Sol.145. (d)

position vectors of vertices A, B and C of triangle ABC are respectively

$\hat{j}+\hat{k}, 3\hat{i}+\hat{j}+5\hat{k}$ and $3\hat{j}+3\hat{k}$.

$\overrightarrow{CA}=2\hat{j}+2\hat{k}, \overrightarrow{CB}=-3\hat{i}+2\hat{j}-2\hat{k}$

angle between above vectors

$\cos\theta=\frac{\vec{d}.\vec{c}}{|\vec{d}||\vec{c}|}=\frac{4-4}{\sqrt{8}\sqrt{17}}=0$

so angle C is 90°

Sol.146. (b)

let $\bar{b}$ is $x\hat{i}+y\hat{j}+z\hat{k}$

$\vec{a}=3\hat{i}+3\hat{j}+3\hat{k}$

$\bar{a}.\bar{b}=27$

$3x+3y+3z=27$

$x+y+z=9$ …….(i)

$\bar{a}\times\bar{b}=9\,\bar{c}$

$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 3 & 3\\ x & y & z\end{vmatrix}=9(\hat{j}-\hat{k})$

$(3z-3y)\hat{i}-(3z-3x)\hat{j}+(3y-3x)\hat{k}=9\hat{j}-9\hat{k}$

$3z-3y=0$

$z=y$ ...…..(ii)

$3z-3x=-9$

$z-x=-3$

$x=z+3$ ……(iii)

by (i), (ii), (iii)

$z=y=2$ and $x=5$

$\vec{b}=5\hat{i}+2\hat{j}+2\hat{k}$

Sol.147. (a)

$\vec{a}=3\hat{i}+3\hat{j}+3\hat{k}$ and $\vec{b}=5\hat{i}+2\hat{j}+2\hat{k}$

$\vec{a}+\vec{b}=8\hat{i}+5\hat{j}+5\hat{k}=\vec{d}$

angle between $\vec{d}$ and $\vec{c}$ is

$\cos\theta=\frac{\vec{d}.\vec{c}}{|\vec{d}||\vec{c}|}=\frac{0+45-45}{\sqrt{114}\sqrt{162}}=0$

so $(\bar{a}+\bar{b})$ and $\bar{c}$ are perpendicular.

Sol.148. (b)

unit vector along the $\bar{a}=4\hat{i}-8\hat{j}+\hat{k}$ is

$\frac{4}{9}\hat{i}-\frac{8}{9}\hat{j}+\frac{1}{9}\hat{k}$

so $\cos\alpha=4/9$

Sol.149. (a)

$\cos 2\beta+\cos 2\gamma=2\cos^2\beta-1+2\cos^2\gamma-1$

$=2\left(\frac{64}{81}\right)-1+2\left(\frac{1}{81}\right)-1=\frac{130}{81}-2=-\frac{32}{81}$

Sol.150. (b)

The position vectors of two points A and B are $\hat{i}-\hat{j}$ and $\hat{j}+\hat{k}$ respectively.

so coordinates of A(1, –1, 0) and B(0, 1, 1) equation of line AB = $\frac{x-1}{-1}=\frac{y+1}{2}=\frac{z}{1}$

points (–1, 3, 2) and (–2, 5, 3) are satisfying above equation so these points lie on line AB.
so answer is (b) option.

Sol.151. (c)
by above solution
distance between A and B
$=\sqrt{(0-1)^2+(1+1)^2+(1-0)^2}=\sqrt{6}$

Sol.152. (b)
projection of the vector $\hat{i}+2\hat{j}+3\hat{k}$ on the vector $2\hat{i}+3\hat{j}-2\hat{k}$

$=\frac{\vec{a}.\vec{b}}{|\vec{b}|}=\frac{(\hat{i}+2\hat{j}+3\hat{k})(2\hat{i}+3\hat{j}-2\hat{k})}{|2\hat{i}+3\hat{j}-2\hat{k}|}=\frac{2}{\sqrt{17}}$

Sol.153. (a)
$(\vec{a}\times\vec{b})^2+(\vec{a}.\vec{b})^2=144$
$a^2b^2\sin^2\theta+a^2b^2\cos^2\theta=144$
$a^2b^2=144$
$a^2(16)=144$
$a=3$

Sol.154. (c)
$\vec{a}.\vec{b}\geq 0$
$ab\cos\theta\geq 0$
$\cos\theta\geq 0$
$0\leq\theta\leq\pi/2$

Sol.155.
position vector of A is $60\hat{i}+3\hat{j}$, so coordinates of A are (60, 3)
position vector of B is $40\hat{i}-8\hat{j}$, so coordinates of B are (40, –8)
position vector of C is $\beta\hat{i}-52\hat{j}$, so coordinates of C are (β, –52)
if A, B, C are collinear then by concept of 2D geometry

$\begin{vmatrix}60 & 3 & 1\\ 40 & -8 & 1\\ \beta & -52 & 1\end{vmatrix}=0$

$\beta=-40$

Sol.156. (b)
cross product of two vectors represents a perpendicular vector of both vectors, and there are always two unit vectors,
first outward to the plane of a and b i.e. $\frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$
and second is inward to the plane of a and b i.e. $-\frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$
so statement 1 is wrong.
angle between $\vec{a}=(0,1,1)$ and $\vec{b}=(1,0,1)$ is

$\cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}|.|\vec{b}|}=\frac{0+0+1}{\sqrt{2}\sqrt{2}}=\frac{1}{2}$

so θ = 60°

Sol.157. (*)
$\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}.\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c}$
so $\vec{a}\times(\vec{b}\times\vec{a})=(\vec{a}.\vec{a})\vec{b}-(\vec{a}.\vec{b})\vec{a}$
$\vec{a}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$
$(\vec{a}.\vec{a})\vec{b}-(\vec{a}.\vec{b})\vec{a}=(3)\vec{b}-(-2)\vec{a}$
$=3(\hat{i}+2\hat{j}-\hat{k})+2(\hat{i}-\hat{j}+\hat{k})=5\hat{i}+4\hat{j}-\hat{k}$
$5\hat{i}+4\hat{j}-\hat{k}=\alpha\hat{i}-\beta\hat{j}+\gamma\hat{k}$
α = 5, β = – 4, γ = – 1
α + β + γ = 0

Sol.158. (a)
A vector makes π/3 angle with $2\hat{i}$ or x – axis
so cosα = 1/2
π/4 angle with $3\hat{j}$ or y – axis so cosβ = 1/√2
we know that $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$
$1/4+1/2+\cos^2\gamma=1$
$\cos^2\gamma=1/4$
cosγ = 1/2
so required vector is
$r(\cos\alpha\hat{i}+\cos\beta\hat{j}+\cos\gamma\hat{k})$
$=2\left(\frac{1}{2}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{2}\hat{k}\right)=\hat{i}+\sqrt{2}\hat{j}+\hat{k}$

Sol.159. (b)
moment of force about a point depends on the position vector from the point to the point of application of the force. statement 1 is incorrect.
torque is a vector quantity. statement 2 is correct.

Sol.160. (a)
$(\vec{r}.\hat{i})(\vec{r}\times\hat{i})+(\vec{r}.\hat{j})(\vec{r}\times\hat{j})+(\vec{r}.\hat{k})(\vec{r}\times\hat{k})$
let r is any vector for example $\hat{i}+\hat{j}+\hat{k}$ or $2\hat{i}+3\hat{j}-\hat{k}$ or $\hat{i}$ or $x\hat{i}+y\hat{j}+z\hat{k}$
for easy calculation we are taking r as $\hat{i}$
$(\hat{i}.\hat{i})(\hat{i}\times\hat{i})+(\hat{i}.\hat{j})(\hat{i}\times\hat{j})+(\hat{i}.\hat{k})(\hat{i}\times\hat{k})$
= 0 + 0 + 0 = 0

Sol.161. (b)
angle between $\vec{a}$ and $\vec{a}-\vec{b}$ is

$\cos\phi=\frac{\vec{a}.(\vec{a}-\vec{b})}{|\vec{a}||\vec{a}-\vec{b}|}=\frac{a^2-\vec{a}.\vec{b}}{a\sqrt{a^2+b^2-2\vec{a}.\vec{b}}}$

$=\frac{a^2-ab\cos\theta}{a\sqrt{a^2+b^2-2ab\cos\theta}}=\frac{16-16\left(\frac{1}{2}\right)}{4\sqrt{16+16-2(4)(4)\left(\frac{1}{2}\right)}}$

$\cos\phi=\frac{8}{4\times 4}=\frac{1}{2}\Rightarrow\varphi=\frac{\pi}{3}$

Sol.162. (a)
$(2\hat{i}+6\hat{j}+27\hat{k})\times(\hat{i}+\alpha\hat{j}+\beta\hat{k})=\vec{0}$

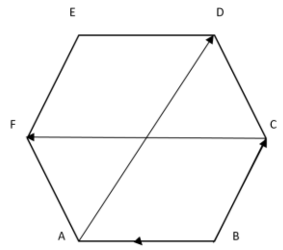
$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 6 & 27\\ 1 & \alpha & \beta\end{vmatrix}=\vec{0}$

$(6\beta-27\alpha)\hat{i}-(2\beta-27)\hat{j}+(2\alpha-6)\hat{k}=\vec{0}$
2α – 6 = 0 so α = 3
2β – 27 = 0 so β = 27/2
3α + 2β = 9 + 27 = 36

Sol.163. (b)
$(\vec{a}\times\vec{b})+\sqrt{3}(\vec{a}.\vec{b})$
$ab\sin\theta+\sqrt{3}ab\cos\theta$
$ab(\sin\theta+\sqrt{3}\cos\theta)$
$2ab\left(\frac{1}{2}\sin\theta+\frac{\sqrt{3}}{2}\cos\theta\right)$
$2ab\left(\cos\frac{\pi}{3}\sin\theta+\cos\frac{\pi}{3}\cos\theta\right)$
$2ab\sin\left(\theta+\frac{\pi}{3}\right)$
maximum value of sine lie at 90° so θ = 30°

Sol.164. (a)
If $\vec{a}+2\vec{b}$ is perpendicular to $5\vec{a}-4\vec{b}$
$(\vec{a}+2\vec{b})(5\vec{a}-4\vec{b})=0$
$(\vec{a}+2\vec{b})(5\vec{a}-4\vec{b})=5a^2+10\vec{b}.\vec{a}-4\vec{a}\vec{b}-8b^2=0$
$5+6ab\cos\theta-8=0$
$\cos\theta=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{3}$
cosθ + cos2θ = cos60 + cos120 = ½ – ½ = 0

Sol.165. (a)



In a regular hexagon we know that
AD = 2BC = 2FE and CF = 2BA = 2DE
Given that
$\overrightarrow{AD}=m\overrightarrow{BC}$ and $\overrightarrow{CF}=n\overrightarrow{AB}$
AD and BC are in same direction so m = 2
CF and AB are in opposite direction so n = – 2
mn = 2(–2) = – 4

Sol.166. (c)
$\vec{a}=\hat{i}+\hat{j}$, magnitude $|\vec{a}|=\sqrt{1+1}=\sqrt{2}$
$\vec{b}=\hat{j}+\hat{k}$, magnitude $|\vec{b}|=\sqrt{1+1}=\sqrt{2}$
angle between $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=\hat{j}+\hat{k}$

$\cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}|.|\vec{b}|}=\frac{0+1+0}{\sqrt{2}\sqrt{2}}=\frac{1}{2}$

θ = 60°
statement 1 :
magnitude of $\hat{i}+\hat{k}=\sqrt{1+1}=\sqrt{2}$
angle between $\vec{a}=\hat{i}+\hat{j}$ and $\hat{i}+\hat{k}$

$\cos\theta=\frac{1+0+0}{\sqrt{2}\sqrt{2}}=\frac{1}{2}$

θ = 60°
statement 1 is correct.
statement 2 :
magnitude of $\frac{(-\hat{i}+4\hat{j}-\hat{k})}{3}=\sqrt{\frac{1}{9}+\frac{16}{9}+\frac{1}{9}}=\sqrt{2}$
angle between $\vec{a}=\hat{i}+\hat{j}$ and $\frac{(-\hat{i}+4\hat{j}-\hat{k})}{3}$

$\cos\theta=\frac{\frac{-1}{3}+\frac{4}{3}+0}{\sqrt{2}\sqrt{2}}=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{3}$

statement 2 is correct.

Sol.167. (d)
$\vec{a}=(\sin^2\alpha)\hat{i}+(\sin^2\beta)\hat{j}+(\sin^2\gamma)\hat{k}$

$\vec b = \hat i + \hat j + \hat k$

$\vec a.\vec b = \sin^2\alpha + \sin^2\beta + \sin^2\gamma$

$= 1-\cos^2\alpha + 1-\cos^2\beta + 1-\cos^2\gamma$

$= 3-(\cos^2\alpha+\cos^2\beta+\cos^2\gamma) = 3-1 = 2$

Sol.168. (c)

$\vec d = \left(\vec a\times\vec b\right)\times\vec c$

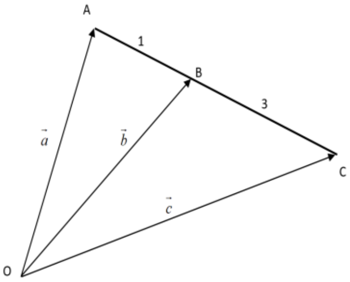
i.e. $\vec d$ is perpendicular to $\left(\vec a\times\vec b\right)$ and $\vec c$, if $\vec d$ is perpendicular to $\left(\vec a\times\vec b\right)$ then $\vec d$ is coplanar with $\vec a$ and $\vec b$.

Sol.169. (b)

$3\vec a - 4\vec b + \vec c = \vec 0$

$\vec b = \frac{3\vec a+\vec c}{4} = \frac{3\vec a+\vec c}{3+1} = \frac{BC\vec a + AB\vec c}{BC+AB}$

Now by diagram



AB : BC = 1 : 3

Sol.170. (a)

$\left(\vec a\times\vec b\right)+\left(\vec b\times\vec c\right)+\left(\vec c\times\vec a\right)$

$\left(\vec a\times\vec b\right)+\left[\vec b\times\left\{\left(\cos^2\theta\right)\vec a+\left(\sin^2\theta\right)\vec b\right\}\right]$

$+\left[\left\{\left(\cos^2\theta\right)\vec a+\left(\sin^2\theta\right)\vec b\right\}\times\vec a\right]$

$\left(\vec a\times\vec b\right)+\left[\left(\cos^2\theta\right)\left(\vec b\times\vec a\right)\right]+\left[\left(\sin^2\theta\right)\left(\vec b\times\vec a\right)\right]$

$\left(\vec a\times\vec b\right)+\left[\left(\cos^2\theta+\sin^2\theta\right)\left(\vec b\times\vec a\right)\right]$

$\left(\vec a\times\vec b\right)+\left(\vec b\times\vec a\right)=\left(\vec a\times\vec b\right)-\left(\vec a\times\vec b\right)=\vec 0$

Sol.171. (a)

$\left|\vec a\times\vec b\right| = ab\sin\theta = 1$

$\sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2}$

$\vec a.\vec b = ab\cos\theta = \cos\frac{\pi}{2} = 0$

Sol.172. (d)

$\vec p\times\vec q = \left(\vec a-\vec b\right)\times\left(\vec a+\vec b\right) = 2\left(\vec a\times\vec b\right)$

We know that $\left|\vec a.\vec b\right|^2 + \left|\vec a\times\vec b\right|^2 = a^2b^2$

$4+\left|\vec a\times\vec b\right|^2 = (4)(4)$

$\left|\vec a\times\vec b\right| = 2\sqrt3$

$\vec p\times\vec q = 2\left(\vec a\times\vec b\right) = 4\sqrt3$

Sol.173. (d)

$\vec a = 2\hat i - \hat j + \hat k$ and $\vec b = \hat i + \hat j + 3\hat k$

$\vec a\times\vec b = \begin{vmatrix}\hat i & \hat j & \hat k\\ 2 & -1 & 1\\ 1 & 1 & 3\end{vmatrix} = -4\hat i - 5\hat j - 3\hat k$

This $\vec a\times\vec b$ is perpendicular to given vectors. And all vectors parallel to this vectors are $\lambda\left(-4\hat i-5\hat j-3\hat k\right)$

Put $\lambda = -1, 2, \frac{1}{50}$, so all given vectors in question are perpendicular to $2\hat i - \hat j + \hat k$ and $\hat i + \hat j + 3\hat k$

Sol.174. (c)

$\vec a = \hat i + 2\hat j + 3\hat k$ and $\vec b = 2\hat i + \hat j + 2\hat k$

$\vec a\times\vec b = \begin{vmatrix}\hat i & \hat j & \hat k\\ 1 & 2 & 3\\ 2 & 1 & 2\end{vmatrix} = \hat i + 4\hat j - 3\hat k$

Area of parallelogram =

$\left|\vec a\times\vec b\right| = \sqrt{1+16+9} = \sqrt{26}$

Sol.175. (a)

Posiion vectors of the vertices A, B, C and D are $3\hat i+4\hat j-2\hat k$, $4\hat i-4\hat j-3\hat k$, $2\hat i-3\hat j+2\hat k$ and $6\hat i-2\hat j+\hat k$ respectively

$\overrightarrow{AC} = -\hat i - 7\hat j + 4\hat k$ and $\overrightarrow{BD} = 2\hat i + 2\hat j + 4\hat k$

Angle between AC and BD = $\frac{\overrightarrow{AC}.\overrightarrow{BD}}{\left|\overrightarrow{AC}\right|\left|\overrightarrow{BD}\right|}$

$= \frac{-2-14+16}{\sqrt{1+49+16}\sqrt{4+4+16}} = 0$

So both are perpendicular

Sol.176. (a)

$\overrightarrow{BA} = 2\hat i + 4\hat j + 2\hat k$

Moment = $\vec r\times\vec F = \begin{vmatrix}\hat i & \hat j & \hat k\\ 2 & 4 & 2\\ 2 & -\lambda & 5\end{vmatrix}$

$= (20+2\lambda)\hat i - 6\hat j + (-2\lambda-8)\hat k$

Now compare this with given moment in question $16\hat i - 6\hat j + 2\lambda\hat k$

$= (20+2\lambda) = 16 \Rightarrow \lambda = -2$